

Recursive Organization Improvement

Supplementary Tables S1–S5

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This arXiv ancillary supplement reports Study 1 (evidence acquisition and retention), Study 2 (review arrangements and evidence generation), and the supplementary audit controls. Study 2 and the acquisition-matched controls are exploratory. Source paths below are relative to the repository root. Every uncertainty entry is a 95% Monte Carlo half-width conditional on the supplied simulation parameters. Rounding to zero does not imply a general absence of uncertainty. Protocol definitions and cost units differ between the studies and are specified in the manuscript. Supplementary table panels retain all rows from the named analyses; no outcome-based row filtering is used.

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Supplementary Table S1: Study 1: evidence acquisition and retention

Source: results/learning_summary.csv; results/learning_contrasts.csv.

Core design: 54 conditions, each with 128 independent replicate trajectories of 48 rounds. Window 8 includes eight previous rounds plus current observations. Net value, regret and expenses are per production task. Harm is the percentage of rounds selecting a template worse than standard. Late net value covers the final eight rounds. Panel C separates core and exploratory control contrasts; intervals use paired replicate differences.

A. Net value, harmful adoption, and selection regret

Environment	Memory	Arm	Net value	Harm (%)	Regret
Stationary	Reset	Biased	0.39666 ± 0.00278	15.755 ± 0.800	0.08514 ± 0.00278
Stationary	Reset	Balanced	0.45224 ± 0.00123	0.537 ± 0.177	0.02939 ± 0.00123
Stationary	Reset	Neyman	0.45389 ± 0.00118	0.521 ± 0.175	0.02781 ± 0.00118
Stationary	Reset	Rejection	0.45400 ± 0.00128	0.505 ± 0.174	0.02795 ± 0.00128
Stationary	Reset	Once	0.43664 ± 0.00476	4.704 ± 1.256	0.04509 ± 0.00476
Stationary	Reset	Repeated	0.43522 ± 0.00261	4.980 ± 0.718	0.04667 ± 0.00261
Stationary	Cumulative	Biased	0.46917 ± 0.00267	1.009 ± 0.376	0.01263 ± 0.00267
Stationary	Cumulative	Balanced	0.48007 ± 0.00050	0.000 ± 0.000	0.00156 ± 0.00050
Stationary	Cumulative	Neyman	0.48046 ± 0.00048	0.000 ± 0.000	0.00133 ± 0.00048
Stationary	Cumulative	Rejection	0.48059 ± 0.00054	0.000 ± 0.000	0.00137 ± 0.00054
Stationary	Cumulative	Once	0.47928 ± 0.00124	0.016 ± 0.032	0.00248 ± 0.00124
Stationary	Cumulative	Repeated	0.48000 ± 0.00080	0.016 ± 0.032	0.00191 ± 0.00080
Stationary	Window 8	Biased	0.45936 ± 0.00284	1.514 ± 0.496	0.02244 ± 0.00284
Stationary	Window 8	Balanced	0.47921 ± 0.00073	0.000 ± 0.000	0.00242 ± 0.00073
Stationary	Window 8	Neyman	0.47962 ± 0.00071	0.000 ± 0.000	0.00217 ± 0.00071
Stationary	Window 8	Rejection	0.48002 ± 0.00069	0.000 ± 0.000	0.00193 ± 0.00069
Stationary	Window 8	Once	0.47537 ± 0.00194	0.309 ± 0.280	0.00639 ± 0.00194
Stationary	Window 8	Repeated	0.47790 ± 0.00118	0.016 ± 0.032	0.00400 ± 0.00118
Reversal	Reset	Biased	0.52595 ± 0.00282	8.203 ± 0.594	0.07584 ± 0.00282
Reversal	Reset	Balanced	0.58280 ± 0.00128	0.244 ± 0.117	0.01883 ± 0.00128
Reversal	Reset	Neyman	0.58300 ± 0.00121	0.244 ± 0.117	0.01870 ± 0.00121
Reversal	Reset	Rejection	0.58435 ± 0.00115	0.244 ± 0.117	0.01760 ± 0.00115
Reversal	Reset	Once	0.56442 ± 0.00519	2.702 ± 0.754	0.03731 ± 0.00520
Reversal	Reset	Repeated	0.56282 ± 0.00253	2.523 ± 0.522	0.03907 ± 0.00253
Reversal	Cumulative	Biased	0.47211 ± 0.00288	1.009 ± 0.376	0.12968 ± 0.00288
Reversal	Cumulative	Balanced	0.48026 ± 0.00060	0.000 ± 0.000	0.12137 ± 0.00060
Reversal	Cumulative	Neyman	0.48054 ± 0.00051	0.000 ± 0.000	0.12125 ± 0.00051
Reversal	Cumulative	Rejection	0.48285 ± 0.00131	0.000 ± 0.000	0.11910 ± 0.00131
Reversal	Cumulative	Once	0.47963 ± 0.00134	0.016 ± 0.032	0.12212 ± 0.00134
Reversal	Cumulative	Repeated	0.48347 ± 0.00235	0.016 ± 0.032	0.11843 ± 0.00235
Reversal	Window 8	Biased	0.56094 ± 0.00291	1.270 ± 0.460	0.04085 ± 0.00291
Reversal	Window 8	Balanced	0.57583 ± 0.00089	0.000 ± 0.000	0.02580 ± 0.00089
Reversal	Window 8	Neyman	0.57601 ± 0.00081	0.000 ± 0.000	0.02578 ± 0.00081
Reversal	Window 8	Rejection	0.57859 ± 0.00096	0.000 ± 0.000	0.02336 ± 0.00096
Reversal	Window 8	Once	0.57231 ± 0.00206	0.212 ± 0.247	0.02945 ± 0.00207
Reversal	Window 8	Repeated	0.58206 ± 0.00231	0.016 ± 0.032	0.01984 ± 0.00231
Uniform gain	Reset	Biased	0.66166 ± 0.00295	0.000 ± 0.000	0.06014 ± 0.00295
Uniform gain	Reset	Balanced	0.71411 ± 0.00111	0.000 ± 0.000	0.00752 ± 0.00111
Uniform gain	Reset	Neyman	0.71317 ± 0.00114	0.000 ± 0.000	0.00852 ± 0.00114
Uniform gain	Reset	Rejection	0.71619 ± 0.00097	0.000 ± 0.000	0.00576 ± 0.00097
Uniform gain	Reset	Once	0.69615 ± 0.00458	0.000 ± 0.000	0.02559 ± 0.00459
Uniform gain	Reset	Repeated	0.69412 ± 0.00234	0.000 ± 0.000	0.02777 ± 0.00234
Uniform gain	Cumulative	Biased	0.71837 ± 0.00126	0.000 ± 0.000	0.00342 ± 0.00126
Uniform gain	Cumulative	Balanced	0.72147 ± 0.00019	0.000 ± 0.000	0.00016 ± 0.00019
Uniform gain	Cumulative	Neyman	0.72159 ± 0.00025	0.000 ± 0.000	0.00020 ± 0.00025
Uniform gain	Cumulative	Rejection	0.72184 ± 0.00017	0.000 ± 0.000	0.00012 ± 0.00017
Uniform gain	Cumulative	Once	0.72173 ± 0.00008	0.000 ± 0.000	0.00004 ± 0.00008
Uniform gain	Cumulative	Repeated	0.72187 ± 0.00008	0.000 ± 0.000	0.00004 ± 0.00008
Uniform gain	Window 8	Biased	0.71662 ± 0.00165	0.000 ± 0.000	0.00518 ± 0.00165
Uniform gain	Window 8	Balanced	0.72147 ± 0.00019	0.000 ± 0.000	0.00016 ± 0.00019

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Environment	Memory	Arm	Net value	Harm (%)	Regret
Uniform gain	Window 8	Neyman	0.72159 ± 0.00025	0.000 ± 0.000	0.00020 ± 0.00025
Uniform gain	Window 8	Rejection	0.72184 ± 0.00017	0.000 ± 0.000	0.00012 ± 0.00017
Uniform gain	Window 8	Once	0.72101 ± 0.00080	0.000 ± 0.000	0.00076 ± 0.00080
Uniform gain	Window 8	Repeated	0.72186 ± 0.00008	0.000 ± 0.000	0.00004 ± 0.00008

B. Evaluation expense, program-trial expense, and late outcome

Environment	Memory	Arm	Expense	Trial expense	Late net
Stationary	Reset	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.39935 ± 0.00766
Stationary	Reset	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.45327 ± 0.00317
Stationary	Reset	Neyman	0.01830 ± 0.00000	0.00000 ± 0.00000	0.45577 ± 0.00305
Stationary	Reset	Rejection	0.01805 ± 0.00000	0.00000 ± 0.00000	0.45670 ± 0.00320
Stationary	Reset	Once	0.01827 ± 0.00001	0.00060 ± 0.00000	0.44309 ± 0.00568
Stationary	Reset	Repeated	0.01811 ± 0.00000	0.00361 ± 0.00000	0.43967 ± 0.00566
Stationary	Cumulative	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.48062 ± 0.00186
Stationary	Cumulative	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.48163 ± 0.00000
Stationary	Cumulative	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.48179 ± 0.00000
Stationary	Cumulative	Rejection	0.01805 ± 0.00000	0.00000 ± 0.00000	0.48195 ± 0.00000
Stationary	Cumulative	Once	0.01824 ± 0.00001	0.00060 ± 0.00000	0.48173 ± 0.00001
Stationary	Cumulative	Repeated	0.01809 ± 0.00000	0.00359 ± 0.00000	0.48191 ± 0.00000
Stationary	Window 8	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.46571 ± 0.00562
Stationary	Window 8	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.47976 ± 0.00216
Stationary	Window 8	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.47992 ± 0.00211
Stationary	Window 8	Rejection	0.01805 ± 0.00000	0.00000 ± 0.00000	0.48066 ± 0.00185
Stationary	Window 8	Once	0.01824 ± 0.00001	0.00060 ± 0.00000	0.47719 ± 0.00265
Stationary	Window 8	Repeated	0.01810 ± 0.00000	0.00359 ± 0.00000	0.47968 ± 0.00185
Reversal	Reset	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.65722 ± 0.00820
Reversal	Reset	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.71237 ± 0.00297
Reversal	Reset	Neyman	0.01830 ± 0.00000	0.00000 ± 0.00000	0.71102 ± 0.00318
Reversal	Reset	Rejection	0.01805 ± 0.00000	0.00000 ± 0.00000	0.71445 ± 0.00250
Reversal	Reset	Once	0.01827 ± 0.00001	0.00060 ± 0.00000	0.69498 ± 0.00676
Reversal	Reset	Repeated	0.01811 ± 0.00000	0.00361 ± 0.00000	0.69142 ± 0.00633
Reversal	Cumulative	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.49808 ± 0.00972
Reversal	Cumulative	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.48280 ± 0.00189
Reversal	Cumulative	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.48226 ± 0.00092
Reversal	Cumulative	Rejection	0.01805 ± 0.00000	0.00000 ± 0.00000	0.49484 ± 0.00631
Reversal	Cumulative	Once	0.01824 ± 0.00001	0.00060 ± 0.00000	0.48384 ± 0.00307
Reversal	Cumulative	Repeated	0.01809 ± 0.00000	0.00359 ± 0.00000	0.49785 ± 0.00945
Reversal	Window 8	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72062 ± 0.00152
Reversal	Window 8	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.72163 ± 0.00000
Reversal	Window 8	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72179 ± 0.00000
Reversal	Window 8	Rejection	0.01805 ± 0.00000	0.00000 ± 0.00000	0.72195 ± 0.00000
Reversal	Window 8	Once	0.01824 ± 0.00001	0.00060 ± 0.00000	0.72056 ± 0.00151
Reversal	Window 8	Repeated	0.01810 ± 0.00000	0.00359 ± 0.00000	0.72190 ± 0.00000
Uniform gain	Reset	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.65722 ± 0.00820
Uniform gain	Reset	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.71237 ± 0.00297
Uniform gain	Reset	Neyman	0.01831 ± 0.00000	0.00000 ± 0.00000	0.71102 ± 0.00318
Uniform gain	Reset	Rejection	0.01805 ± 0.00000	0.00000 ± 0.00000	0.71445 ± 0.00250
Uniform gain	Reset	Once	0.01826 ± 0.00001	0.00060 ± 0.00000	0.69077 ± 0.00763
Uniform gain	Reset	Repeated	0.01811 ± 0.00000	0.00361 ± 0.00000	0.69142 ± 0.00633
Uniform gain	Cumulative	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72179 ± 0.00000
Uniform gain	Cumulative	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.72163 ± 0.00000
Uniform gain	Cumulative	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72179 ± 0.00000
Uniform gain	Cumulative	Rejection	0.01805 ± 0.00000	0.00000 ± 0.00000	0.72195 ± 0.00000
Uniform gain	Cumulative	Once	0.01823 ± 0.00001	0.00060 ± 0.00000	0.72175 ± 0.00001
Uniform gain	Cumulative	Repeated	0.01809 ± 0.00000	0.00359 ± 0.00000	0.72191 ± 0.00000
Uniform gain	Window 8	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72062 ± 0.00152
Uniform gain	Window 8	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.72163 ± 0.00000
Uniform gain	Window 8	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72179 ± 0.00000
Uniform gain	Window 8	Rejection	0.01805 ± 0.00000	0.00000 ± 0.00000	0.72195 ± 0.00000

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Environment	Memory	Arm	Expense	Trial expense	Late net
Uniform gain	Window 8	Once	0.01823 ± 0.00001	0.00060 ± 0.00000	0.72175 ± 0.00001
Uniform gain	Window 8	Repeated	0.01810 ± 0.00000	0.00359 ± 0.00000	0.72190 ± 0.00000

Reversal labels in Panel C give the one-based round before which the change occurs.

C. Paired net-value contrasts

Study	Environment	Memory	Comparison	<i>N</i>	Difference
core	Stationary	Reset	Repeated - Balanced	128	-0.01702 ± 0.00262
core	Stationary	Reset	Repeated - Successive rejects	128	-0.01878 ± 0.00266
core	Stationary	Reset	Repeated - Discover once	128	-0.00142 ± 0.00441
core	Stationary	Reset	Repeated - uniform fixed mixture	128	0.00096 ± 0.00252
core	Stationary	Cumulative	Repeated - Balanced	128	-0.00007 ± 0.00092
core	Stationary	Cumulative	Repeated - Successive rejects	128	-0.00059 ± 0.00097
core	Stationary	Cumulative	Repeated - Discover once	128	0.00071 ± 0.00070
core	Stationary	Cumulative	Repeated - uniform fixed mixture	128	0.00343 ± 0.00117
core	Stationary	Window 8	Repeated - Balanced	128	-0.00130 ± 0.00114
core	Stationary	Window 8	Repeated - Successive rejects	128	-0.00212 ± 0.00115
core	Stationary	Window 8	Repeated - Discover once	128	0.00254 ± 0.00166
core	Stationary	Window 8	Repeated - uniform fixed mixture	128	0.00518 ± 0.00137
core	Stationary	Cumulative	Balanced - Reset	128	0.02783 ± 0.00124
core	Stationary	Window 8	Balanced - Reset	128	0.02697 ± 0.00126
core	Stationary	Cumulative	Repeated discovery - Reset	128	0.04478 ± 0.00283
core	Stationary	Window 8	Repeated discovery - Reset	128	0.04268 ± 0.00291
core	Stationary	Cumulative	Discover once - Reset	128	0.04264 ± 0.00456
core	Stationary	Window 8	Discover once - Reset	128	0.03873 ± 0.00364
core	Reversal at 25	Reset	Repeated - Balanced	128	-0.01997 ± 0.00235
core	Reversal at 25	Reset	Repeated - Successive rejects	128	-0.02153 ± 0.00241
core	Reversal at 25	Reset	Repeated - Discover once	128	-0.00159 ± 0.00500
core	Reversal at 25	Reset	Repeated - uniform fixed mixture	128	-0.00109 ± 0.00224
core	Reversal at 25	Cumulative	Repeated - Balanced	128	0.00321 ± 0.00245
core	Reversal at 25	Cumulative	Repeated - Successive rejects	128	0.00062 ± 0.00273
core	Reversal at 25	Cumulative	Repeated - Discover once	128	0.00384 ± 0.00250
core	Reversal at 25	Cumulative	Repeated - uniform fixed mixture	128	0.00583 ± 0.00259
core	Reversal at 25	Window 8	Repeated - Balanced	128	0.00623 ± 0.00240
core	Reversal at 25	Window 8	Repeated - Successive rejects	128	0.00347 ± 0.00252
core	Reversal at 25	Window 8	Repeated - Discover once	128	0.00975 ± 0.00267
core	Reversal at 25	Window 8	Repeated - uniform fixed mixture	128	0.01114 ± 0.00255
core	Reversal at 25	Cumulative	Balanced - Reset	128	-0.10253 ± 0.00132
core	Reversal at 25	Window 8	Balanced - Reset	128	-0.00697 ± 0.00136
core	Reversal at 25	Cumulative	Repeated discovery - Reset	128	-0.07935 ± 0.00345
core	Reversal at 25	Window 8	Repeated discovery - Reset	128	0.01924 ± 0.00351
core	Reversal at 25	Cumulative	Discover once - Reset	128	-0.08478 ± 0.00493
core	Reversal at 25	Window 8	Discover once - Reset	128	0.00789 ± 0.00423
core	Uniform gain	Reset	Repeated - Balanced	128	-0.02000 ± 0.00240
core	Uniform gain	Reset	Repeated - Successive rejects	128	-0.02208 ± 0.00239
core	Uniform gain	Reset	Repeated - Discover once	128	-0.00204 ± 0.00464
core	Uniform gain	Reset	Repeated - uniform fixed mixture	128	-0.00220 ± 0.00223
core	Uniform gain	Cumulative	Repeated - Balanced	128	0.00039 ± 0.00013
core	Uniform gain	Cumulative	Repeated - Successive rejects	128	0.00003 ± 0.00011
core	Uniform gain	Cumulative	Repeated - Discover once	128	0.00013 ± 0.00001
core	Uniform gain	Cumulative	Repeated - uniform fixed mixture	128	0.00139 ± 0.00044
core	Uniform gain	Window 8	Repeated - Balanced	128	0.00039 ± 0.00013
core	Uniform gain	Window 8	Repeated - Successive rejects	128	0.00003 ± 0.00011
core	Uniform gain	Window 8	Repeated - Discover once	128	0.00085 ± 0.00080
core	Uniform gain	Window 8	Repeated - uniform fixed mixture	128	0.00197 ± 0.00056
core	Uniform gain	Cumulative	Balanced - Reset	128	0.00736 ± 0.00110
core	Uniform gain	Window 8	Balanced - Reset	128	0.00736 ± 0.00110
core	Uniform gain	Cumulative	Repeated discovery - Reset	128	0.02775 ± 0.00235
core	Uniform gain	Window 8	Repeated discovery - Reset	128	0.02775 ± 0.00235

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Study	Environment	Memory	Comparison	N	Difference
core	Uniform gain	Cumulative	Discover once - Reset	128	0.02558 ± 0.00459
core	Uniform gain	Window 8	Discover once - Reset	128	0.02486 ± 0.00437
exploratory control	Stationary	Reset	Repeated - Balanced	128	-0.01665 ± 0.00255
exploratory control	Stationary	Reset	Repeated - Trial-matched Balanced	128	-0.01382 ± 0.00238
exploratory control	Stationary	Reset	Repeated - Discover once	128	0.00035 ± 0.00448
exploratory control	Stationary	Cumulative	Repeated - Balanced	128	0.00078 ± 0.00091
exploratory control	Stationary	Cumulative	Repeated - Trial-matched Balanced	128	-0.00016 ± 0.00046
exploratory control	Stationary	Cumulative	Repeated - Discover once	128	0.00110 ± 0.00135
exploratory control	Stationary	Window 8	Repeated - Balanced	128	0.00023 ± 0.00104
exploratory control	Stationary	Window 8	Repeated - Trial-matched Balanced	128	-0.00116 ± 0.00058
exploratory control	Stationary	Window 8	Repeated - Discover once	128	0.00538 ± 0.00210
exploratory control	Reversal at 25	Reset	Repeated - Balanced	128	-0.01808 ± 0.00239
exploratory control	Reversal at 25	Reset	Repeated - Trial-matched Balanced	128	-0.01525 ± 0.00224
exploratory control	Reversal at 25	Reset	Repeated - Discover once	128	-0.00223 ± 0.00438
exploratory control	Reversal at 25	Cumulative	Repeated - Balanced	128	0.00922 ± 0.00353
exploratory control	Reversal at 25	Cumulative	Repeated - Trial-matched Balanced	128	0.00832 ± 0.00363
exploratory control	Reversal at 25	Cumulative	Repeated - Discover once	128	0.00926 ± 0.00397
exploratory control	Reversal at 25	Window 8	Repeated - Balanced	128	0.00607 ± 0.00217
exploratory control	Reversal at 25	Window 8	Repeated - Trial-matched Balanced	128	0.00191 ± 0.00207
exploratory control	Reversal at 25	Window 8	Repeated - Discover once	128	0.00889 ± 0.00220
exploratory control	Reversal at 21	Reset	Repeated - Balanced	128	-0.01817 ± 0.00254
exploratory control	Reversal at 21	Reset	Repeated - Trial-matched Balanced	128	-0.01586 ± 0.00228
exploratory control	Reversal at 21	Reset	Repeated - Discover once	128	-0.00242 ± 0.00439
exploratory control	Reversal at 21	Cumulative	Repeated - Balanced	128	0.00840 ± 0.00513
exploratory control	Reversal at 21	Cumulative	Repeated - Trial-matched Balanced	128	0.01132 ± 0.00526
exploratory control	Reversal at 21	Cumulative	Repeated - Discover once	128	0.00891 ± 0.00550
exploratory control	Reversal at 21	Window 8	Repeated - Balanced	128	0.00031 ± 0.00118
exploratory control	Reversal at 21	Window 8	Repeated - Trial-matched Balanced	128	-0.00063 ± 0.00072
exploratory control	Reversal at 21	Window 8	Repeated - Discover once	128	0.00257 ± 0.00137
exploratory control	Reversal at 29	Reset	Repeated - Balanced	128	-0.01700 ± 0.00252
exploratory control	Reversal at 29	Reset	Repeated - Trial-matched Balanced	128	-0.01399 ± 0.00235
exploratory control	Reversal at 29	Reset	Repeated - Discover once	128	-0.00116 ± 0.00433
exploratory control	Reversal at 29	Cumulative	Repeated - Balanced	128	0.00262 ± 0.00148
exploratory control	Reversal at 29	Cumulative	Repeated - Trial-matched Balanced	128	0.00168 ± 0.00136
exploratory control	Reversal at 29	Cumulative	Repeated - Discover once	128	0.00293 ± 0.00225
exploratory control	Reversal at 29	Window 8	Repeated - Balanced	128	0.00004 ± 0.00120
exploratory control	Reversal at 29	Window 8	Repeated - Trial-matched Balanced	128	-0.00057 ± 0.00069
exploratory control	Reversal at 29	Window 8	Repeated - Discover once	128	0.00342 ± 0.00177

Supplementary Table S2: Retained programs and conditional block outcomes

Source: `results/learning_conditional.csv`.

All 189 observed program groups are retained: three environments and three memories, both discover-once and repeated-discovery rules, and all three retained programs at their scheduled reviews. Shares use 128 runs per review condition; n is the number retaining the named program. Net and harm cover the subsequent eight-round block. Harm is expressed as a percentage. These are descriptive, endogenously selected groups; no causal program effect or unavailable conditional interval is implied. Every reported group has at least one retained run.

Complete conditional summary

Environment	Memory	Arm	Round	Program	n	Share (%)	Net	Harm (%)
Stationary	Reset	Once	1	Biased	35	27.34	0.39126	17.857
Stationary	Reset	Once	1	Balanced	48	37.50	0.44462	0.781
Stationary	Reset	Once	1	Neyman	45	35.16	0.44897	1.389
Stationary	Reset	Repeated	1	Biased	35	27.34	0.39126	17.857
Stationary	Reset	Repeated	1	Balanced	48	37.50	0.44462	0.781
Stationary	Reset	Repeated	1	Neyman	45	35.16	0.44897	1.389
Stationary	Reset	Repeated	9	Biased	33	25.78	0.39189	18.182
Stationary	Reset	Repeated	9	Balanced	38	29.69	0.45446	0.658
Stationary	Reset	Repeated	9	Neyman	57	44.53	0.44748	1.096
Stationary	Reset	Repeated	17	Biased	40	31.25	0.40474	11.562
Stationary	Reset	Repeated	17	Balanced	40	31.25	0.44701	1.250
Stationary	Reset	Repeated	17	Neyman	48	37.50	0.45291	0.260
Stationary	Reset	Repeated	25	Biased	42	32.81	0.40127	15.774
Stationary	Reset	Repeated	25	Balanced	40	31.25	0.45433	0.625
Stationary	Reset	Repeated	25	Neyman	46	35.94	0.45002	1.359
Stationary	Reset	Repeated	33	Biased	39	30.47	0.40016	14.744
Stationary	Reset	Repeated	33	Balanced	41	32.03	0.44695	1.829
Stationary	Reset	Repeated	33	Neyman	48	37.50	0.44908	1.042
Stationary	Reset	Repeated	41	Biased	35	27.34	0.40549	11.071
Stationary	Reset	Repeated	41	Balanced	55	42.97	0.45203	0.455
Stationary	Reset	Repeated	41	Neyman	38	29.69	0.45328	0.658
Stationary	Cumulative	Once	1	Biased	35	27.34	0.46315	0.357
Stationary	Cumulative	Once	1	Balanced	48	37.50	0.47470	0.000
Stationary	Cumulative	Once	1	Neyman	45	35.16	0.47489	0.000
Stationary	Cumulative	Repeated	1	Biased	35	27.34	0.46315	0.357
Stationary	Cumulative	Repeated	1	Balanced	48	37.50	0.47470	0.000
Stationary	Cumulative	Repeated	1	Neyman	45	35.16	0.47489	0.000
Stationary	Cumulative	Repeated	9	Biased	34	26.56	0.48146	0.000
Stationary	Cumulative	Repeated	9	Balanced	46	35.94	0.48191	0.000
Stationary	Cumulative	Repeated	9	Neyman	48	37.50	0.48160	0.000
Stationary	Cumulative	Repeated	17	Biased	43	33.59	0.47912	0.000
Stationary	Cumulative	Repeated	17	Balanced	44	34.38	0.48191	0.000
Stationary	Cumulative	Repeated	17	Neyman	41	32.03	0.48191	0.000
Stationary	Cumulative	Repeated	25	Biased	43	33.59	0.48191	0.000
Stationary	Cumulative	Repeated	25	Balanced	46	35.94	0.48191	0.000
Stationary	Cumulative	Repeated	25	Neyman	39	30.47	0.48191	0.000
Stationary	Cumulative	Repeated	33	Biased	46	35.94	0.48191	0.000
Stationary	Cumulative	Repeated	33	Balanced	41	32.03	0.48191	0.000
Stationary	Cumulative	Repeated	33	Neyman	41	32.03	0.48191	0.000
Stationary	Cumulative	Repeated	41	Biased	49	38.28	0.48191	0.000
Stationary	Cumulative	Repeated	41	Balanced	40	31.25	0.48191	0.000
Stationary	Cumulative	Repeated	41	Neyman	39	30.47	0.48191	0.000
Stationary	Window 8	Once	1	Biased	35	27.34	0.46315	0.357
Stationary	Window 8	Once	1	Balanced	48	37.50	0.47470	0.000
Stationary	Window 8	Once	1	Neyman	45	35.16	0.47489	0.000
Stationary	Window 8	Repeated	1	Biased	35	27.34	0.46315	0.357
Stationary	Window 8	Repeated	1	Balanced	48	37.50	0.47470	0.000

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Environment	Memory	Arm	Round	Program	n	Share (%)	Net	Harm (%)
Stationary	Window 8	Repeated	1	Neyman	45	35.16	0.47489	0.000
Stationary	Window 8	Repeated	9	Biased	34	26.56	0.47838	0.000
Stationary	Window 8	Repeated	9	Balanced	46	35.94	0.47832	0.000
Stationary	Window 8	Repeated	9	Neyman	48	37.50	0.48128	0.000
Stationary	Window 8	Repeated	17	Biased	56	43.75	0.47628	0.000
Stationary	Window 8	Repeated	17	Balanced	35	27.34	0.48190	0.000
Stationary	Window 8	Repeated	17	Neyman	37	28.91	0.48069	0.000
Stationary	Window 8	Repeated	25	Biased	51	39.84	0.47808	0.000
Stationary	Window 8	Repeated	25	Balanced	39	30.47	0.48191	0.000
Stationary	Window 8	Repeated	25	Neyman	38	29.69	0.47875	0.000
Stationary	Window 8	Repeated	33	Biased	50	39.06	0.47591	0.000
Stationary	Window 8	Repeated	33	Balanced	34	26.56	0.47882	0.000
Stationary	Window 8	Repeated	33	Neyman	44	34.38	0.48020	0.000
Stationary	Window 8	Repeated	41	Biased	41	32.03	0.47678	0.000
Stationary	Window 8	Repeated	41	Balanced	38	29.69	0.48032	0.000
Stationary	Window 8	Repeated	41	Neyman	49	38.28	0.48160	0.000
Reversal	Reset	Once	1	Biased	35	27.34	0.39126	17.857
Reversal	Reset	Once	1	Balanced	48	37.50	0.44462	0.781
Reversal	Reset	Once	1	Neyman	45	35.16	0.44897	1.389
Reversal	Reset	Repeated	1	Biased	35	27.34	0.39126	17.857
Reversal	Reset	Repeated	1	Balanced	48	37.50	0.44462	0.781
Reversal	Reset	Repeated	1	Neyman	45	35.16	0.44897	1.389
Reversal	Reset	Repeated	9	Biased	33	25.78	0.39189	18.182
Reversal	Reset	Repeated	9	Balanced	38	29.69	0.45446	0.658
Reversal	Reset	Repeated	9	Neyman	57	44.53	0.44748	1.096
Reversal	Reset	Repeated	17	Biased	40	31.25	0.40474	11.562
Reversal	Reset	Repeated	17	Balanced	40	31.25	0.44701	1.250
Reversal	Reset	Repeated	17	Neyman	48	37.50	0.45291	0.260
Reversal	Reset	Repeated	25	Biased	47	36.72	0.66508	0.000
Reversal	Reset	Repeated	25	Balanced	45	35.16	0.71022	0.000
Reversal	Reset	Repeated	25	Neyman	36	28.12	0.70606	0.000
Reversal	Reset	Repeated	33	Biased	58	45.31	0.66603	0.000
Reversal	Reset	Repeated	33	Balanced	29	22.66	0.71206	0.000
Reversal	Reset	Repeated	33	Neyman	41	32.03	0.70908	0.000
Reversal	Reset	Repeated	41	Biased	61	47.66	0.67001	0.000
Reversal	Reset	Repeated	41	Balanced	29	22.66	0.71258	0.000
Reversal	Reset	Repeated	41	Neyman	38	29.69	0.70965	0.000
Reversal	Cumulative	Once	1	Biased	35	27.34	0.46315	0.357
Reversal	Cumulative	Once	1	Balanced	48	37.50	0.47470	0.000
Reversal	Cumulative	Once	1	Neyman	45	35.16	0.47489	0.000
Reversal	Cumulative	Repeated	1	Biased	35	27.34	0.46315	0.357
Reversal	Cumulative	Repeated	1	Balanced	48	37.50	0.47470	0.000
Reversal	Cumulative	Repeated	1	Neyman	45	35.16	0.47489	0.000
Reversal	Cumulative	Repeated	9	Biased	34	26.56	0.48146	0.000
Reversal	Cumulative	Repeated	9	Balanced	46	35.94	0.48191	0.000
Reversal	Cumulative	Repeated	9	Neyman	48	37.50	0.48160	0.000
Reversal	Cumulative	Repeated	17	Biased	43	33.59	0.47912	0.000
Reversal	Cumulative	Repeated	17	Balanced	44	34.38	0.48191	0.000
Reversal	Cumulative	Repeated	17	Neyman	41	32.03	0.48191	0.000
Reversal	Cumulative	Repeated	25	Biased	43	33.59	0.48191	0.000
Reversal	Cumulative	Repeated	25	Balanced	46	35.94	0.48452	0.000
Reversal	Cumulative	Repeated	25	Neyman	39	30.47	0.48191	0.000
Reversal	Cumulative	Repeated	33	Biased	45	35.16	0.48191	0.000
Reversal	Cumulative	Repeated	33	Balanced	42	32.81	0.49405	0.000
Reversal	Cumulative	Repeated	33	Neyman	41	32.03	0.48191	0.000
Reversal	Cumulative	Repeated	41	Biased	49	38.28	0.49171	0.000
Reversal	Cumulative	Repeated	41	Balanced	40	31.25	0.50816	0.000
Reversal	Cumulative	Repeated	41	Neyman	39	30.47	0.49499	0.000
Reversal	Window 8	Once	1	Biased	35	27.34	0.46315	0.357
Reversal	Window 8	Once	1	Balanced	48	37.50	0.47470	0.000
Reversal	Window 8	Once	1	Neyman	45	35.16	0.47489	0.000
Reversal	Window 8	Repeated	1	Biased	35	27.34	0.46315	0.357
Reversal	Window 8	Repeated	1	Balanced	48	37.50	0.47470	0.000
Reversal	Window 8	Repeated	1	Neyman	45	35.16	0.47489	0.000

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Environment	Memory	Arm	Round	Program	n	Share (%)	Net	Harm (%)
Reversal	Window 8	Repeated	9	Biased	34	26.56	0.47838	0.000
Reversal	Window 8	Repeated	9	Balanced	46	35.94	0.47832	0.000
Reversal	Window 8	Repeated	9	Neyman	48	37.50	0.48128	0.000
Reversal	Window 8	Repeated	17	Biased	56	43.75	0.47628	0.000
Reversal	Window 8	Repeated	17	Balanced	35	27.34	0.48190	0.000
Reversal	Window 8	Repeated	17	Neyman	37	28.91	0.48069	0.000
Reversal	Window 8	Repeated	25	Biased	51	39.84	0.60573	0.000
Reversal	Window 8	Repeated	25	Balanced	39	30.47	0.61883	0.000
Reversal	Window 8	Repeated	25	Neyman	38	29.69	0.63506	0.000
Reversal	Window 8	Repeated	33	Biased	56	43.75	0.72191	0.000
Reversal	Window 8	Repeated	33	Balanced	36	28.12	0.72191	0.000
Reversal	Window 8	Repeated	33	Neyman	36	28.12	0.72191	0.000
Reversal	Window 8	Repeated	41	Biased	61	47.66	0.72191	0.000
Reversal	Window 8	Repeated	41	Balanced	26	20.31	0.72190	0.000
Reversal	Window 8	Repeated	41	Neyman	41	32.03	0.72190	0.000
Uniform gain	Reset	Once	1	Biased	46	35.94	0.67592	0.000
Uniform gain	Reset	Once	1	Balanced	35	27.34	0.71374	0.000
Uniform gain	Reset	Once	1	Neyman	47	36.72	0.71295	0.000
Uniform gain	Reset	Repeated	1	Biased	46	35.94	0.67592	0.000
Uniform gain	Reset	Repeated	1	Balanced	35	27.34	0.71374	0.000
Uniform gain	Reset	Repeated	1	Neyman	47	36.72	0.71295	0.000
Uniform gain	Reset	Repeated	9	Biased	44	34.38	0.66939	0.000
Uniform gain	Reset	Repeated	9	Balanced	35	27.34	0.70860	0.000
Uniform gain	Reset	Repeated	9	Neyman	49	38.28	0.71148	0.000
Uniform gain	Reset	Repeated	17	Biased	46	35.94	0.67069	0.000
Uniform gain	Reset	Repeated	17	Balanced	39	30.47	0.70727	0.000
Uniform gain	Reset	Repeated	17	Neyman	43	33.59	0.70829	0.000
Uniform gain	Reset	Repeated	25	Biased	47	36.72	0.66508	0.000
Uniform gain	Reset	Repeated	25	Balanced	45	35.16	0.71022	0.000
Uniform gain	Reset	Repeated	25	Neyman	36	28.12	0.70606	0.000
Uniform gain	Reset	Repeated	33	Biased	58	45.31	0.66603	0.000
Uniform gain	Reset	Repeated	33	Balanced	29	22.66	0.71206	0.000
Uniform gain	Reset	Repeated	33	Neyman	41	32.03	0.70908	0.000
Uniform gain	Reset	Repeated	41	Biased	61	47.66	0.67001	0.000
Uniform gain	Reset	Repeated	41	Balanced	29	22.66	0.71258	0.000
Uniform gain	Reset	Repeated	41	Neyman	38	29.69	0.70965	0.000
Uniform gain	Cumulative	Once	1	Biased	46	35.94	0.72125	0.000
Uniform gain	Cumulative	Once	1	Balanced	35	27.34	0.72188	0.000
Uniform gain	Cumulative	Once	1	Neyman	47	36.72	0.72189	0.000
Uniform gain	Cumulative	Repeated	1	Biased	46	35.94	0.72125	0.000
Uniform gain	Cumulative	Repeated	1	Balanced	35	27.34	0.72188	0.000
Uniform gain	Cumulative	Repeated	1	Neyman	47	36.72	0.72189	0.000
Uniform gain	Cumulative	Repeated	9	Biased	34	26.56	0.72191	0.000
Uniform gain	Cumulative	Repeated	9	Balanced	39	30.47	0.72191	0.000
Uniform gain	Cumulative	Repeated	9	Neyman	55	42.97	0.72191	0.000
Uniform gain	Cumulative	Repeated	17	Biased	33	25.78	0.72191	0.000
Uniform gain	Cumulative	Repeated	17	Balanced	38	29.69	0.72191	0.000
Uniform gain	Cumulative	Repeated	17	Neyman	57	44.53	0.72191	0.000
Uniform gain	Cumulative	Repeated	25	Biased	36	28.12	0.72191	0.000
Uniform gain	Cumulative	Repeated	25	Balanced	39	30.47	0.72191	0.000
Uniform gain	Cumulative	Repeated	25	Neyman	53	41.41	0.72191	0.000
Uniform gain	Cumulative	Repeated	33	Biased	36	28.12	0.72191	0.000
Uniform gain	Cumulative	Repeated	33	Balanced	36	28.12	0.72191	0.000
Uniform gain	Cumulative	Repeated	33	Neyman	56	43.75	0.72191	0.000
Uniform gain	Cumulative	Repeated	41	Biased	39	30.47	0.72191	0.000
Uniform gain	Cumulative	Repeated	41	Balanced	33	25.78	0.72191	0.000
Uniform gain	Cumulative	Repeated	41	Neyman	56	43.75	0.72191	0.000
Uniform gain	Window 8	Once	1	Biased	46	35.94	0.72125	0.000
Uniform gain	Window 8	Once	1	Balanced	35	27.34	0.72188	0.000
Uniform gain	Window 8	Once	1	Neyman	47	36.72	0.72189	0.000
Uniform gain	Window 8	Repeated	1	Biased	46	35.94	0.72125	0.000
Uniform gain	Window 8	Repeated	1	Balanced	35	27.34	0.72188	0.000
Uniform gain	Window 8	Repeated	1	Neyman	47	36.72	0.72189	0.000
Uniform gain	Window 8	Repeated	9	Biased	34	26.56	0.72191	0.000

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Environment	Memory	Arm	Round	Program	n	Share (%)	Net	Harm (%)
Uniform gain	Window 8	Repeated	9	Balanced	39	30.47	0.72191	0.000
Uniform gain	Window 8	Repeated	9	Neyman	55	42.97	0.72191	0.000
Uniform gain	Window 8	Repeated	17	Biased	41	32.03	0.72190	0.000
Uniform gain	Window 8	Repeated	17	Balanced	30	23.44	0.72191	0.000
Uniform gain	Window 8	Repeated	17	Neyman	57	44.53	0.72191	0.000
Uniform gain	Window 8	Repeated	25	Biased	42	32.81	0.72191	0.000
Uniform gain	Window 8	Repeated	25	Balanced	38	29.69	0.72190	0.000
Uniform gain	Window 8	Repeated	25	Neyman	48	37.50	0.72191	0.000
Uniform gain	Window 8	Repeated	33	Biased	52	40.62	0.72190	0.000
Uniform gain	Window 8	Repeated	33	Balanced	34	26.56	0.72191	0.000
Uniform gain	Window 8	Repeated	33	Neyman	42	32.81	0.72191	0.000
Uniform gain	Window 8	Repeated	41	Biased	61	47.66	0.72191	0.000
Uniform gain	Window 8	Repeated	41	Balanced	26	20.31	0.72190	0.000
Uniform gain	Window 8	Repeated	41	Neyman	41	32.03	0.72190	0.000

Supplementary Table S3: Audit policy and memory-window grid

Source: `results/robustness_summary.csv`.

All 54 combinations of three environments, three memory windows, and six audit policies. Each cell uses 200 independent seeds. Paired differences compare with fixed audit rate .10 within the same environment and window. The outcome is net value per task under the supplementary audit model.

Complete audit grid

Environment	Window	Policy	Net value	Difference vs. fixed .10
Stationary	6	No audit	0.88054 ± 0.00117	0.02452 ± 0.00080
Stationary	6	Fixed 0.05	0.85918 ± 0.00181	0.00316 ± 0.00119
Stationary	6	Fixed 0.10	0.85602 ± 0.00152	0.00000 ± 0.00000
Stationary	6	Fixed 0.40	0.80061 ± 0.00120	-0.05541 ± 0.00078
Stationary	6	Periodic dense	0.84527 ± 0.00148	-0.01075 ± 0.00108
Stationary	6	Adaptive audit	0.85356 ± 0.00135	-0.00246 ± 0.00082
Stationary	12	No audit	0.88054 ± 0.00117	0.02118 ± 0.00047
Stationary	12	Fixed 0.05	0.86517 ± 0.00177	0.00580 ± 0.00109
Stationary	12	Fixed 0.10	0.85937 ± 0.00130	0.00000 ± 0.00000
Stationary	12	Fixed 0.40	0.80066 ± 0.00119	-0.05871 ± 0.00049
Stationary	12	Periodic dense	0.85290 ± 0.00120	-0.00647 ± 0.00051
Stationary	12	Adaptive audit	0.86681 ± 0.00127	0.00744 ± 0.00060
Stationary	24	No audit	0.88054 ± 0.00117	0.02037 ± 0.00027
Stationary	24	Fixed 0.05	0.86869 ± 0.00147	0.00851 ± 0.00067
Stationary	24	Fixed 0.10	0.86018 ± 0.00120	0.00000 ± 0.00000
Stationary	24	Fixed 0.40	0.80066 ± 0.00119	-0.05952 ± 0.00031
Stationary	24	Periodic dense	0.85319 ± 0.00118	-0.00698 ± 0.00028
Stationary	24	Adaptive audit	0.87162 ± 0.00125	0.01144 ± 0.00037
Slow shifts	6	No audit	0.56603 ± 0.00195	-0.09302 ± 0.00284
Slow shifts	6	Fixed 0.05	0.64384 ± 0.00319	-0.01521 ± 0.00231
Slow shifts	6	Fixed 0.10	0.65905 ± 0.00249	0.00000 ± 0.00000
Slow shifts	6	Fixed 0.40	0.62772 ± 0.00153	-0.03133 ± 0.00222
Slow shifts	6	Periodic dense	0.64667 ± 0.00239	-0.01238 ± 0.00245
Slow shifts	6	Adaptive audit	0.66438 ± 0.00198	0.00533 ± 0.00234
Slow shifts	12	No audit	0.56603 ± 0.00195	-0.09364 ± 0.00277
Slow shifts	12	Fixed 0.05	0.65240 ± 0.00322	-0.00727 ± 0.00266
Slow shifts	12	Fixed 0.10	0.65967 ± 0.00241	0.00000 ± 0.00000
Slow shifts	12	Fixed 0.40	0.61216 ± 0.00148	-0.04751 ± 0.00202
Slow shifts	12	Periodic dense	0.66029 ± 0.00204	0.00062 ± 0.00236
Slow shifts	12	Adaptive audit	0.66785 ± 0.00239	0.00818 ± 0.00263
Slow shifts	24	No audit	0.56603 ± 0.00195	-0.06813 ± 0.00271
Slow shifts	24	Fixed 0.05	0.63665 ± 0.00326	0.00249 ± 0.00266
Slow shifts	24	Fixed 0.10	0.63416 ± 0.00242	0.00000 ± 0.00000
Slow shifts	24	Fixed 0.40	0.57706 ± 0.00157	-0.05710 ± 0.00205
Slow shifts	24	Periodic dense	0.63149 ± 0.00278	-0.00267 ± 0.00300
Slow shifts	24	Adaptive audit	0.65398 ± 0.00290	0.01982 ± 0.00316
Fast shifts	6	No audit	0.56512 ± 0.00210	-0.02974 ± 0.00259
Fast shifts	6	Fixed 0.05	0.59498 ± 0.00279	0.00012 ± 0.00226
Fast shifts	6	Fixed 0.10	0.59486 ± 0.00251	0.00000 ± 0.00000
Fast shifts	6	Fixed 0.40	0.54718 ± 0.00180	-0.04768 ± 0.00209
Fast shifts	6	Periodic dense	0.58839 ± 0.00256	-0.00647 ± 0.00247
Fast shifts	6	Adaptive audit	0.59544 ± 0.00234	0.00058 ± 0.00247
Fast shifts	12	No audit	0.56512 ± 0.00210	0.03871 ± 0.00253
Fast shifts	12	Fixed 0.05	0.54272 ± 0.00307	0.01632 ± 0.00243
Fast shifts	12	Fixed 0.10	0.52640 ± 0.00295	0.00000 ± 0.00000
Fast shifts	12	Fixed 0.40	0.45976 ± 0.00214	-0.06665 ± 0.00221
Fast shifts	12	Periodic dense	0.52383 ± 0.00234	-0.00257 ± 0.00265
Fast shifts	12	Adaptive audit	0.52467 ± 0.00273	-0.00173 ± 0.00289
Fast shifts	24	No audit	0.56512 ± 0.00210	-0.01550 ± 0.00239
Fast shifts	24	Fixed 0.05	0.58156 ± 0.00266	0.00093 ± 0.00242
Fast shifts	24	Fixed 0.10	0.58062 ± 0.00237	0.00000 ± 0.00000
Fast shifts	24	Fixed 0.40	0.54112 ± 0.00239	-0.03950 ± 0.00270
Fast shifts	24	Periodic dense	0.57131 ± 0.00240	-0.00931 ± 0.00244
Fast shifts	24	Adaptive audit	0.56229 ± 0.00220	-0.01834 ± 0.00251

Supplementary Table S4: Expected-value-of-information sensitivity

Source: results/audit_extensions_summary.csv.

All 72 combinations of three priors, two routing thresholds, two candidate-generation costs, two environments, and three policies. Each cell uses 96 independent test seeds; development seeds selected the horizon separately. Nonzero generation cost charges counterfactual A outputs when route B is executed. Paired differences use fixed audit rate .10 within the same condition.

Complete prior, threshold, and generation-cost grid

Prior	Threshold	g_c	Environment	Policy	Net value	Difference vs. fixed .10
1:1	0.14	0	Stationary	Fixed 0.10	0.84794 $\pm$ 0.00236	0.00000 $\pm$ 0.00000
1:1	0.14	0	Stationary	Adaptive audit	0.85300 $\pm$ 0.00213	0.00506 $\pm$ 0.00173
1:1	0.14	0	Stationary	EVSI	0.83379 $\pm$ 0.00359	-0.01415 $\pm$ 0.00326
1:1	0.14	0	Regime shifts	Fixed 0.10	0.66159 $\pm$ 0.00308	0.00000 $\pm$ 0.00000
1:1	0.14	0	Regime shifts	Adaptive audit	0.67534 $\pm$ 0.00293	0.01375 $\pm$ 0.00317
1:1	0.14	0	Regime shifts	EVSI	0.67584 $\pm$ 0.00569	0.01426 $\pm$ 0.00540
1:1	0.14	0.2	Stationary	Fixed 0.10	0.84718 $\pm$ 0.00244	0.00000 $\pm$ 0.00000
1:1	0.14	0.2	Stationary	Adaptive audit	0.85173 $\pm$ 0.00220	0.00455 $\pm$ 0.00184
1:1	0.14	0.2	Stationary	EVSI	0.82623 $\pm$ 0.00362	-0.02094 $\pm$ 0.00338
1:1	0.14	0.2	Regime shifts	Fixed 0.10	0.65182 $\pm$ 0.00305	0.00000 $\pm$ 0.00000
1:1	0.14	0.2	Regime shifts	Adaptive audit	0.66587 $\pm$ 0.00299	0.01405 $\pm$ 0.00323
1:1	0.14	0.2	Regime shifts	EVSI	0.66778 $\pm$ 0.00568	0.01596 $\pm$ 0.00542
1:1	0.16	0	Stationary	Fixed 0.10	0.85228 $\pm$ 0.00191	0.00000 $\pm$ 0.00000
1:1	0.16	0	Stationary	Adaptive audit	0.85706 $\pm$ 0.00194	0.00479 $\pm$ 0.00134
1:1	0.16	0	Stationary	EVSI	0.83421 $\pm$ 0.00355	-0.01807 $\pm$ 0.00313
1:1	0.16	0	Regime shifts	Fixed 0.10	0.65968 $\pm$ 0.00334	0.00000 $\pm$ 0.00000
1:1	0.16	0	Regime shifts	Adaptive audit	0.67286 $\pm$ 0.00277	0.01318 $\pm$ 0.00371
1:1	0.16	0	Regime shifts	EVSI	0.67570 $\pm$ 0.00566	0.01602 $\pm$ 0.00547
1:1	0.16	0.2	Stationary	Fixed 0.10	0.85180 $\pm$ 0.00195	0.00000 $\pm$ 0.00000
1:1	0.16	0.2	Stationary	Adaptive audit	0.85607 $\pm$ 0.00199	0.00427 $\pm$ 0.00145
1:1	0.16	0.2	Stationary	EVSI	0.82674 $\pm$ 0.00357	-0.02506 $\pm$ 0.00321
1:1	0.16	0.2	Regime shifts	Fixed 0.10	0.65064 $\pm$ 0.00327	0.00000 $\pm$ 0.00000
1:1	0.16	0.2	Regime shifts	Adaptive audit	0.66208 $\pm$ 0.00286	0.01143 $\pm$ 0.00367
1:1	0.16	0.2	Regime shifts	EVSI	0.66745 $\pm$ 0.00563	0.01681 $\pm$ 0.00541
1:9	0.14	0	Stationary	Fixed 0.10	0.85814 $\pm$ 0.00172	0.00000 $\pm$ 0.00000
1:9	0.14	0	Stationary	Adaptive audit	0.86494 $\pm$ 0.00192	0.00680 $\pm$ 0.00087
1:9	0.14	0	Stationary	EVSI	0.86557 $\pm$ 0.00207	0.00743 $\pm$ 0.00147
1:9	0.14	0	Regime shifts	Fixed 0.10	0.66200 $\pm$ 0.00333	0.00000 $\pm$ 0.00000
1:9	0.14	0	Regime shifts	Adaptive audit	0.67056 $\pm$ 0.00285	0.00856 $\pm$ 0.00352
1:9	0.14	0	Regime shifts	EVSI	0.65987 $\pm$ 0.00606	-0.00213 $\pm$ 0.00667
1:9	0.14	0.2	Stationary	Fixed 0.10	0.85805 $\pm$ 0.00173	0.00000 $\pm$ 0.00000
1:9	0.14	0.2	Stationary	Adaptive audit	0.86480 $\pm$ 0.00195	0.00675 $\pm$ 0.00094
1:9	0.14	0.2	Stationary	EVSI	0.86507 $\pm$ 0.00222	0.00702 $\pm$ 0.00164
1:9	0.14	0.2	Regime shifts	Fixed 0.10	0.65364 $\pm$ 0.00323	0.00000 $\pm$ 0.00000
1:9	0.14	0.2	Regime shifts	Adaptive audit	0.65984 $\pm$ 0.00290	0.00619 $\pm$ 0.00345
1:9	0.14	0.2	Regime shifts	EVSI	0.65465 $\pm$ 0.00638	0.00100 $\pm$ 0.00689
1:9	0.16	0	Stationary	Fixed 0.10	0.85918 $\pm$ 0.00162	0.00000 $\pm$ 0.00000
1:9	0.16	0	Stationary	Adaptive audit	0.86861 $\pm$ 0.00183	0.00943 $\pm$ 0.00053
1:9	0.16	0	Stationary	EVSI	0.86603 $\pm$ 0.00210	0.00685 $\pm$ 0.00134
1:9	0.16	0	Regime shifts	Fixed 0.10	0.65150 $\pm$ 0.00390	0.00000 $\pm$ 0.00000
1:9	0.16	0	Regime shifts	Adaptive audit	0.66416 $\pm$ 0.00362	0.01266 $\pm$ 0.00468
1:9	0.16	0	Regime shifts	EVSI	0.65263 $\pm$ 0.00621	0.00114 $\pm$ 0.00705
1:9	0.16	0.2	Stationary	Fixed 0.10	0.85915 $\pm$ 0.00162	0.00000 $\pm$ 0.00000
1:9	0.16	0.2	Stationary	Adaptive audit	0.86858 $\pm$ 0.00184	0.00943 $\pm$ 0.00055
1:9	0.16	0.2	Stationary	EVSI	0.86603 $\pm$ 0.00210	0.00687 $\pm$ 0.00135
1:9	0.16	0.2	Regime shifts	Fixed 0.10	0.64420 $\pm$ 0.00377	0.00000 $\pm$ 0.00000
1:9	0.16	0.2	Regime shifts	Adaptive audit	0.65145 $\pm$ 0.00364	0.00725 $\pm$ 0.00449
1:9	0.16	0.2	Regime shifts	EVSI	0.65263 $\pm$ 0.00621	0.00844 $\pm$ 0.00697
1:19	0.14	0	Stationary	Fixed 0.10	0.85951 $\pm$ 0.00162	0.00000 $\pm$ 0.00000
1:19	0.14	0	Stationary	Adaptive audit	0.87199 $\pm$ 0.00177	0.01248 $\pm$ 0.00036

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Prior	Threshold	g_c	Environment	Policy	Net value	Difference vs. fixed .10
1:19	0.14	0	Stationary	EVSI	0.87950 ± 0.00163	0.02000 ± 0.00020
1:19	0.14	0	Regime shifts	Fixed 0.10	0.64678 ± 0.00420	0.00000 ± 0.00000
1:19	0.14	0	Regime shifts	Adaptive audit	0.65715 ± 0.00397	0.01037 ± 0.00451
1:19	0.14	0	Regime shifts	EVSI	0.56729 ± 0.00321	-0.07949 ± 0.00502
1:19	0.14	0.2	Stationary	Fixed 0.10	0.85951 ± 0.00162	0.00000 ± 0.00000
1:19	0.14	0.2	Stationary	Adaptive audit	0.87198 ± 0.00177	0.01247 ± 0.00037
1:19	0.14	0.2	Stationary	EVSI	0.87950 ± 0.00163	0.02000 ± 0.00020
1:19	0.14	0.2	Regime shifts	Fixed 0.10	0.63993 ± 0.00406	0.00000 ± 0.00000
1:19	0.14	0.2	Regime shifts	Adaptive audit	0.64531 ± 0.00381	0.00539 ± 0.00416
1:19	0.14	0.2	Regime shifts	EVSI	0.56729 ± 0.00321	-0.07263 ± 0.00487
1:19	0.16	0	Stationary	Fixed 0.10	0.85954 ± 0.00163	0.00000 ± 0.00000
1:19	0.16	0	Stationary	Adaptive audit	0.87348 ± 0.00167	0.01394 ± 0.00026
1:19	0.16	0	Stationary	EVSI	0.87950 ± 0.00163	0.01997 ± 0.00019
1:19	0.16	0	Regime shifts	Fixed 0.10	0.62630 ± 0.00474	0.00000 ± 0.00000
1:19	0.16	0	Regime shifts	Adaptive audit	0.63964 ± 0.00445	0.01334 ± 0.00580
1:19	0.16	0	Regime shifts	EVSI	0.56729 ± 0.00321	-0.05901 ± 0.00535
1:19	0.16	0.2	Stationary	Fixed 0.10	0.85954 ± 0.00163	0.00000 ± 0.00000
1:19	0.16	0.2	Stationary	Adaptive audit	0.87348 ± 0.00167	0.01394 ± 0.00026
1:19	0.16	0.2	Stationary	EVSI	0.87950 ± 0.00163	0.01997 ± 0.00019
1:19	0.16	0.2	Regime shifts	Fixed 0.10	0.62102 ± 0.00454	0.00000 ± 0.00000
1:19	0.16	0.2	Regime shifts	Adaptive audit	0.62680 ± 0.00414	0.00578 ± 0.00529
1:19	0.16	0.2	Regime shifts	EVSI	0.56729 ± 0.00321	-0.05373 ± 0.00515

Supplementary Table S5: Study 2: review arrangements and evidence generation

Source: `study2/results/learning_summary.csv`; `study2/results/paired_contrasts.csv`.

All 84 environment-policy-memory conditions: six environments, four paid policies crossed with three memories, and two fixed baselines per environment. Each learning condition uses 128 trajectories of 48 rounds; late outcomes cover the final 16 rounds. Error and optimal-choice hit are percentages. Regret is deployment loss above the finite-catalog minimum, excluding acquisition and switching. Acquisition cost is per round; production cost is per task in human-call units. Fixed baselines have deterministic expected-outcome scores; their displayed zero half-widths describe the simulation, not real-world certainty. Paired, Randomized, Omit echo and Final only denote the four observation policies defined in Appendix E.

A. Full-horizon selection and net value

Environment	Policy	Memory	Error (%)	Net value	Hit (%)
No echo	Final only	Reset	10.472 ± 0.056	0.62353 ± 0.00174	0.03 ± 0.04
No echo	Final only	Cumulative	4.919 ± 0.197	0.78045 ± 0.00563	14.47 ± 3.55
No echo	Final only	Window 8	7.116 ± 0.137	0.71668 ± 0.00390	6.38 ± 0.76
No echo	Paired	Reset	1.864 ± 0.018	0.85928 ± 0.00056	33.06 ± 1.16
No echo	Paired	Cumulative	1.772 ± 0.002	0.86475 ± 0.00008	98.70 ± 0.18
No echo	Paired	Window 8	1.772 ± 0.002	0.86475 ± 0.00008	98.70 ± 0.18
No echo	Randomized	Reset	1.944 ± 0.022	0.85786 ± 0.00068	50.85 ± 1.21
No echo	Randomized	Cumulative	1.772 ± 0.003	0.86333 ± 0.00026	56.67 ± 6.18
No echo	Randomized	Window 8	1.772 ± 0.003	0.86321 ± 0.00016	53.99 ± 3.14
No echo	Omit echo	Reset	1.857 ± 0.018	0.86217 ± 0.00056	98.36 ± 0.33
No echo	Omit echo	Cumulative	1.772 ± 0.002	0.86480 ± 0.00008	99.97 ± 0.04
No echo	Omit echo	Window 8	1.772 ± 0.002	0.86480 ± 0.00008	99.97 ± 0.04
No echo	Fixed exposed	Fixed	3.520 ± 0.000	0.87440 ± 0.00000	0.00 ± 0.00
No echo	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
High echo	Final only	Reset	11.109 ± 0.058	0.60564 ± 0.00184	0.00 ± 0.00
High echo	Final only	Cumulative	5.983 ± 0.217	0.74760 ± 0.00640	10.89 ± 3.09
High echo	Final only	Window 8	8.788 ± 0.124	0.66901 ± 0.00365	2.70 ± 0.50
High echo	Paired	Reset	1.880 ± 0.017	0.85753 ± 0.00052	97.92 ± 0.31
High echo	Paired	Cumulative	1.773 ± 0.003	0.86077 ± 0.00009	99.95 ± 0.05
High echo	Paired	Window 8	1.773 ± 0.003	0.86077 ± 0.00009	99.95 ± 0.05
High echo	Randomized	Reset	2.267 ± 0.045	0.84636 ± 0.00134	92.53 ± 0.64
High echo	Randomized	Cumulative	1.782 ± 0.008	0.86078 ± 0.00024	99.82 ± 0.12
High echo	Randomized	Window 8	1.782 ± 0.008	0.86078 ± 0.00024	99.82 ± 0.12
High echo	Omit echo	Reset	9.961 ± 0.023	0.61912 ± 0.00070	0.00 ± 0.00
High echo	Omit echo	Cumulative	9.825 ± 0.004	0.62323 ± 0.00013	0.00 ± 0.00
High echo	Omit echo	Window 8	9.825 ± 0.004	0.62323 ± 0.00013	0.00 ± 0.00
High echo	Fixed exposed	Fixed	9.637 ± 0.000	0.69089 ± 0.00000	0.00 ± 0.00
High echo	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
Both dependences	Final only	Reset	11.045 ± 0.054	0.60834 ± 0.00170	0.00 ± 0.00
Both dependences	Final only	Cumulative	9.728 ± 0.136	0.63630 ± 0.00404	5.49 ± 2.55
Both dependences	Final only	Window 8	10.295 ± 0.079	0.62548 ± 0.00251	1.01 ± 0.27
Both dependences	Paired	Reset	7.822 ± 0.017	0.68223 ± 0.00036	51.84 ± 1.28
Both dependences	Paired	Cumulative	7.423 ± 0.049	0.69099 ± 0.00105	82.29 ± 3.92
Both dependences	Paired	Window 8	7.563 ± 0.034	0.68789 ± 0.00073	70.72 ± 2.72
Both dependences	Randomized	Reset	8.265 ± 0.041	0.66972 ± 0.00115	46.96 ± 1.21
Both dependences	Randomized	Cumulative	7.498 ± 0.063	0.68945 ± 0.00135	77.10 ± 4.85
Both dependences	Randomized	Window 8	7.631 ± 0.040	0.68650 ± 0.00087	66.28 ± 3.03
Both dependences	Omit echo	Reset	13.749 ± 0.098	0.50789 ± 0.00302	0.00 ± 0.00
Both dependences	Omit echo	Cumulative	15.939 ± 0.317	0.43965 ± 0.00969	0.00 ± 0.00
Both dependences	Omit echo	Window 8	15.125 ± 0.212	0.46465 ± 0.00649	0.00 ± 0.00
Both dependences	Fixed exposed	Fixed	9.919 ± 0.000	0.68242 ± 0.00000	0.00 ± 0.00
Both dependences	Fixed blind	Fixed	8.560 ± 0.000	0.72020 ± 0.00000	0.00 ± 0.00
Echo shift	Final only	Reset	10.722 ± 0.054	0.61665 ± 0.00170	0.00 ± 0.00
Echo shift	Final only	Cumulative	5.648 ± 0.217	0.75770 ± 0.00621	11.20 ± 2.78
Echo shift	Final only	Window 8	7.826 ± 0.118	0.69639 ± 0.00341	4.82 ± 0.57

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Environment	Policy	Memory	Error (%)	Net value	Hit (%)
Echo shift	Paired	Reset	1.886 ± 0.020	0.85798 ± 0.00060	65.32 ± 0.88
Echo shift	Paired	Cumulative	1.779 ± 0.006	0.86255 ± 0.00018	98.54 ± 0.18
Echo shift	Paired	Window 8	1.772 ± 0.002	0.86274 ± 0.00008	98.62 ± 0.17
Echo shift	Randomized	Reset	2.148 ± 0.036	0.85081 ± 0.00108	71.22 ± 0.95
Echo shift	Randomized	Cumulative	2.113 ± 0.083	0.85206 ± 0.00238	72.40 ± 2.66
Echo shift	Randomized	Window 8	1.960 ± 0.040	0.85659 ± 0.00117	74.54 ± 2.11
Echo shift	Omit echo	Reset	5.926 ± 0.023	0.74011 ± 0.00071	48.96 ± 0.26
Echo shift	Omit echo	Cumulative	5.797 ± 0.002	0.74404 ± 0.00008	49.97 ± 0.04
Echo shift	Omit echo	Window 8	5.797 ± 0.002	0.74404 ± 0.00008	49.97 ± 0.04
Echo shift	Fixed exposed	Fixed	6.579 ± 0.000	0.78264 ± 0.00000	0.00 ± 0.00
Echo shift	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
Source asymmetry	Final only	Reset	11.009 ± 0.058	0.60849 ± 0.00181	0.00 ± 0.00
Source asymmetry	Final only	Cumulative	6.352 ± 0.191	0.73785 ± 0.00535	5.34 ± 2.06
Source asymmetry	Final only	Window 8	8.247 ± 0.134	0.68513 ± 0.00385	1.84 ± 0.41
Source asymmetry	Paired	Reset	2.983 ± 0.028	0.82450 ± 0.00084	90.67 ± 0.77
Source asymmetry	Paired	Cumulative	2.693 ± 0.004	0.83318 ± 0.00013	99.76 ± 0.13
Source asymmetry	Paired	Window 8	2.693 ± 0.004	0.83318 ± 0.00013	99.76 ± 0.13
Source asymmetry	Randomized	Reset	3.414 ± 0.035	0.81334 ± 0.00107	48.91 ± 1.38
Source asymmetry	Randomized	Cumulative	2.936 ± 0.046	0.82728 ± 0.00116	70.91 ± 5.42
Source asymmetry	Randomized	Window 8	3.009 ± 0.027	0.82545 ± 0.00069	62.27 ± 3.18
Source asymmetry	Omit echo	Reset	4.386 ± 0.094	0.78638 ± 0.00281	0.00 ± 0.00
Source asymmetry	Omit echo	Cumulative	3.546 ± 0.015	0.81160 ± 0.00044	0.00 ± 0.00
Source asymmetry	Omit echo	Window 8	3.546 ± 0.015	0.81160 ± 0.00044	0.00 ± 0.00
Source asymmetry	Fixed exposed	Fixed	5.101 ± 0.000	0.82697 ± 0.00000	0.00 ± 0.00
Source asymmetry	Fixed blind	Fixed	4.240 ± 0.000	0.84980 ± 0.00000	0.00 ± 0.00
Stronger humans	Final only	Reset	18.205 ± 0.073	0.39097 ± 0.00207	0.00 ± 0.00
Stronger humans	Final only	Cumulative	9.522 ± 0.263	0.63271 ± 0.00748	10.04 ± 3.28
Stronger humans	Final only	Window 8	12.956 ± 0.201	0.53738 ± 0.00577	3.79 ± 0.79
Stronger humans	Paired	Reset	5.413 ± 0.040	0.74875 ± 0.00121	77.65 ± 1.07
Stronger humans	Paired	Cumulative	4.623 ± 0.009	0.77306 ± 0.00028	99.37 ± 0.25
Stronger humans	Paired	Window 8	4.628 ± 0.011	0.77289 ± 0.00032	99.20 ± 0.29
Stronger humans	Randomized	Reset	5.629 ± 0.044	0.74241 ± 0.00133	72.31 ± 1.12
Stronger humans	Randomized	Cumulative	4.639 ± 0.014	0.77286 ± 0.00041	98.96 ± 0.35
Stronger humans	Randomized	Window 8	4.646 ± 0.014	0.77262 ± 0.00043	98.73 ± 0.38
Stronger humans	Omit echo	Reset	10.693 ± 0.101	0.59515 ± 0.00304	0.00 ± 0.00
Stronger humans	Omit echo	Cumulative	8.969 ± 0.024	0.64781 ± 0.00072	0.00 ± 0.00
Stronger humans	Omit echo	Window 8	8.982 ± 0.027	0.64741 ± 0.00082	0.00 ± 0.00
Stronger humans	Fixed exposed	Fixed	19.570 ± 0.000	0.39290 ± 0.00000	0.00 ± 0.00
Stronger humans	Fixed blind	Fixed	12.320 ± 0.000	0.60740 ± 0.00000	0.00 ± 0.00

B. Full-horizon costs and regret

Environment	Policy	Memory	Acquisition cost	Production cost	Regret
No echo	Final only	Reset	214.788 ± 0.076	0.4879 ± 0.0049	0.24580 ± 0.00174
No echo	Final only	Cumulative	214.788 ± 0.076	0.9764 ± 0.0341	0.08897 ± 0.00563
No echo	Final only	Window 8	214.788 ± 0.076	0.8658 ± 0.0193	0.15270 ± 0.00390
No echo	Paired	Reset	233.600 ± 0.000	1.3853 ± 0.0024	0.00552 ± 0.00056
No echo	Paired	Cumulative	233.600 ± 0.000	1.2526 ± 0.0004	0.00011 ± 0.00008
No echo	Paired	Window 8	233.600 ± 0.000	1.2526 ± 0.0004	0.00011 ± 0.00008
No echo	Randomized	Reset	232.500 ± 0.000	1.3501 ± 0.0028	0.00720 ± 0.00068
No echo	Randomized	Cumulative	232.500 ± 0.000	1.3367 ± 0.0124	0.00179 ± 0.00026
No echo	Randomized	Window 8	232.500 ± 0.000	1.3420 ± 0.0063	0.00189 ± 0.00016
No echo	Omit echo	Reset	233.600 ± 0.000	1.2538 ± 0.0010	0.00268 ± 0.00056
No echo	Omit echo	Cumulative	233.600 ± 0.000	1.2501 ± 0.0001	0.00005 ± 0.00008
No echo	Omit echo	Window 8	233.600 ± 0.000	1.2501 ± 0.0001	0.00005 ± 0.00008
No echo	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.04749 ± 0.00000
No echo	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.05049 ± 0.00000
High echo	Final only	Reset	209.381 ± 0.065	0.4931 ± 0.0055	0.26102 ± 0.00184
High echo	Final only	Cumulative	209.381 ± 0.065	1.0879 ± 0.0388	0.11915 ± 0.00640
High echo	Final only	Window 8	209.381 ± 0.065	0.8080 ± 0.0206	0.19770 ± 0.00365

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Environment	Policy	Memory	Acquisition cost	Production cost	Regret
High echo	Paired	Reset	233.600 $\pm$ 0.000	1.4516 $\pm$ 0.0006	0.00332 $\pm$ 0.00052
High echo	Paired	Cumulative	233.600 $\pm$ 0.000	1.4501 $\pm$ 0.0001	0.00008 $\pm$ 0.00009
High echo	Paired	Window 8	233.600 $\pm$ 0.000	1.4501 $\pm$ 0.0001	0.00008 $\pm$ 0.00009
High echo	Randomized	Reset	232.500 $\pm$ 0.000	1.4428 $\pm$ 0.0016	0.01475 $\pm$ 0.00134
High echo	Randomized	Cumulative	232.500 $\pm$ 0.000	1.4498 $\pm$ 0.0003	0.00035 $\pm$ 0.00024
High echo	Randomized	Window 8	232.500 $\pm$ 0.000	1.4498 $\pm$ 0.0003	0.00035 $\pm$ 0.00024
High echo	Omit echo	Reset	233.600 $\pm$ 0.000	1.2507 $\pm$ 0.0003	0.24173 $\pm$ 0.00070
High echo	Omit echo	Cumulative	233.600 $\pm$ 0.000	1.2500 $\pm$ 0.0000	0.23763 $\pm$ 0.00013
High echo	Omit echo	Window 8	233.600 $\pm$ 0.000	1.2500 $\pm$ 0.0000	0.23763 $\pm$ 0.00013
High echo	Fixed exposed	Fixed	0.000 $\pm$ 0.000	1.0000 $\pm$ 0.0000	0.22700 $\pm$ 0.00000
High echo	Fixed blind	Fixed	0.000 $\pm$ 0.000	1.1500 $\pm$ 0.0000	0.04649 $\pm$ 0.00000
Both dependences	Final only	Reset	206.524 $\pm$ 0.051	0.4882 $\pm$ 0.0051	0.09392 $\pm$ 0.00170
Both dependences	Final only	Cumulative	206.524 $\pm$ 0.051	1.0708 $\pm$ 0.0502	0.06605 $\pm$ 0.00404
Both dependences	Final only	Window 8	206.524 $\pm$ 0.051	0.7599 $\pm$ 0.0200	0.07684 $\pm$ 0.00251
Both dependences	Paired	Reset	233.600 $\pm$ 0.000	1.3006 $\pm$ 0.0085	0.01346 $\pm$ 0.00036
Both dependences	Paired	Cumulative	233.600 $\pm$ 0.000	1.4639 $\pm$ 0.0232	0.00477 $\pm$ 0.00105
Both dependences	Paired	Window 8	233.600 $\pm$ 0.000	1.4090 $\pm$ 0.0160	0.00786 $\pm$ 0.00073
Both dependences	Randomized	Reset	232.500 $\pm$ 0.000	1.2746 $\pm$ 0.0091	0.02623 $\pm$ 0.00115
Both dependences	Randomized	Cumulative	232.500 $\pm$ 0.000	1.4414 $\pm$ 0.0294	0.00658 $\pm$ 0.00135
Both dependences	Randomized	Window 8	232.500 $\pm$ 0.000	1.3894 $\pm$ 0.0180	0.00951 $\pm$ 0.00087
Both dependences	Omit echo	Reset	233.600 $\pm$ 0.000	1.1266 $\pm$ 0.0064	0.18780 $\pm$ 0.00302
Both dependences	Omit echo	Cumulative	233.600 $\pm$ 0.000	1.2569 $\pm$ 0.0123	0.25611 $\pm$ 0.00969
Both dependences	Omit echo	Window 8	233.600 $\pm$ 0.000	1.2274 $\pm$ 0.0097	0.23109 $\pm$ 0.00649
Both dependences	Fixed exposed	Fixed	0.000 $\pm$ 0.000	1.0000 $\pm$ 0.0000	0.07038 $\pm$ 0.00000
Both dependences	Fixed blind	Fixed	0.000 $\pm$ 0.000	1.1500 $\pm$ 0.0000	0.03260 $\pm$ 0.00000
Echo shift	Final only	Reset	212.034 $\pm$ 0.072	0.4900 $\pm$ 0.0051	0.25136 $\pm$ 0.00170
Echo shift	Final only	Cumulative	212.034 $\pm$ 0.072	1.0534 $\pm$ 0.0368	0.11040 $\pm$ 0.00621
Echo shift	Final only	Window 8	212.034 $\pm$ 0.072	0.8499 $\pm$ 0.0191	0.17166 $\pm$ 0.00341
Echo shift	Paired	Reset	233.600 $\pm$ 0.000	1.4192 $\pm$ 0.0018	0.00484 $\pm$ 0.00060
Echo shift	Paired	Cumulative	233.600 $\pm$ 0.000	1.3526 $\pm$ 0.0004	0.00030 $\pm$ 0.00018
Echo shift	Paired	Window 8	233.600 $\pm$ 0.000	1.3528 $\pm$ 0.0004	0.00011 $\pm$ 0.00008
Echo shift	Randomized	Reset	232.500 $\pm$ 0.000	1.3970 $\pm$ 0.0023	0.01227 $\pm$ 0.00108
Echo shift	Randomized	Cumulative	232.500 $\pm$ 0.000	1.3888 $\pm$ 0.0077	0.01105 $\pm$ 0.00238
Echo shift	Randomized	Window 8	232.500 $\pm$ 0.000	1.3922 $\pm$ 0.0045	0.00652 $\pm$ 0.00117
Echo shift	Omit echo	Reset	233.600 $\pm$ 0.000	1.2530 $\pm$ 0.0008	0.12274 $\pm$ 0.00071
Echo shift	Omit echo	Cumulative	233.600 $\pm$ 0.000	1.2501 $\pm$ 0.0001	0.11881 $\pm$ 0.00008
Echo shift	Omit echo	Window 8	233.600 $\pm$ 0.000	1.2501 $\pm$ 0.0001	0.11881 $\pm$ 0.00008
Echo shift	Fixed exposed	Fixed	0.000 $\pm$ 0.000	1.0000 $\pm$ 0.0000	0.13724 $\pm$ 0.00000
Echo shift	Fixed blind	Fixed	0.000 $\pm$ 0.000	1.1500 $\pm$ 0.0000	0.04849 $\pm$ 0.00000
Source asymmetry	Final only	Reset	210.669 $\pm$ 0.070	0.4852 $\pm$ 0.0054	0.23042 $\pm$ 0.00181
Source asymmetry	Final only	Cumulative	210.669 $\pm$ 0.070	1.0058 $\pm$ 0.0389	0.10114 $\pm$ 0.00535
Source asymmetry	Final only	Window 8	210.669 $\pm$ 0.070	0.7979 $\pm$ 0.0204	0.15384 $\pm$ 0.00385
Source asymmetry	Paired	Reset	233.600 $\pm$ 0.000	1.4483 $\pm$ 0.0014	0.00890 $\pm$ 0.00084
Source asymmetry	Paired	Cumulative	233.600 $\pm$ 0.000	1.4500 $\pm$ 0.0002	0.00024 $\pm$ 0.00013
Source asymmetry	Paired	Window 8	233.600 $\pm$ 0.000	1.4500 $\pm$ 0.0002	0.00024 $\pm$ 0.00013
Source asymmetry	Randomized	Reset	232.500 $\pm$ 0.000	1.3701 $\pm$ 0.0030	0.02028 $\pm$ 0.00107
Source asymmetry	Randomized	Cumulative	232.500 $\pm$ 0.000	1.3925 $\pm$ 0.0108	0.00640 $\pm$ 0.00116
Source asymmetry	Randomized	Window 8	232.500 $\pm$ 0.000	1.3752 $\pm$ 0.0064	0.00822 $\pm$ 0.00069
Source asymmetry	Omit echo	Reset	233.600 $\pm$ 0.000	1.2504 $\pm$ 0.0006	0.04703 $\pm$ 0.00281
Source asymmetry	Omit echo	Cumulative	233.600 $\pm$ 0.000	1.2500 $\pm$ 0.0000	0.02183 $\pm$ 0.00044
Source asymmetry	Omit echo	Window 8	233.600 $\pm$ 0.000	1.2500 $\pm$ 0.0000	0.02183 $\pm$ 0.00044
Source asymmetry	Fixed exposed	Fixed	0.000 $\pm$ 0.000	1.0000 $\pm$ 0.0000	0.06349 $\pm$ 0.00000
Source asymmetry	Fixed blind	Fixed	0.000 $\pm$ 0.000	1.1500 $\pm$ 0.0000	0.04066 $\pm$ 0.00000
Stronger humans	Final only	Reset	211.370 $\pm$ 0.066	0.5579 $\pm$ 0.0063	0.38811 $\pm$ 0.00207
Stronger humans	Final only	Cumulative	211.370 $\pm$ 0.066	1.4993 $\pm$ 0.0418	0.14645 $\pm$ 0.00748
Stronger humans	Final only	Window 8	211.370 $\pm$ 0.066	1.1133 $\pm$ 0.0188	0.24175 $\pm$ 0.00576
Stronger humans	Paired	Reset	233.600 $\pm$ 0.000	1.5894 $\pm$ 0.0041	0.02498 $\pm$ 0.00121
Stronger humans	Paired	Cumulative	233.600 $\pm$ 0.000	1.5611 $\pm$ 0.0008	0.00070 $\pm$ 0.00028
Stronger humans	Paired	Window 8	233.600 $\pm$ 0.000	1.5614 $\pm$ 0.0010	0.00088 $\pm$ 0.00032
Stronger humans	Randomized	Reset	232.500 $\pm$ 0.000	1.5957 $\pm$ 0.0047	0.03157 $\pm$ 0.00133
Stronger humans	Randomized	Cumulative	232.500 $\pm$ 0.000	1.5609 $\pm$ 0.0009	0.00118 $\pm$ 0.00041
Stronger humans	Randomized	Window 8	232.500 $\pm$ 0.000	1.5618 $\pm$ 0.0013	0.00142 $\pm$ 0.00043
Stronger humans	Omit echo	Reset	233.600 $\pm$ 0.000	1.3492 $\pm$ 0.0046	0.17857 $\pm$ 0.00304
Stronger humans	Omit echo	Cumulative	233.600 $\pm$ 0.000	1.3039 $\pm$ 0.0012	0.12595 $\pm$ 0.00072

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Environment	Policy	Memory	Acquisition cost	Production cost	Regret
Stronger humans	Omit echo	Window 8	233.600 ± 0.000	1.3053 ± 0.0015	0.12636 ± 0.00082
Stronger humans	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.43790 ± 0.00000
Stronger humans	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.22340 ± 0.00000

C. Final-16-round outcomes

Environment	Policy	Memory	Late error (%)	Late net	Late hit (%)
No echo	Final only	Reset	10.445 ± 0.097	0.62531 ± 0.00303	0.05 ± 0.10
No echo	Final only	Cumulative	3.130 ± 0.228	0.83006 ± 0.00638	27.78 ± 6.96
No echo	Final only	Window 8	7.290 ± 0.265	0.71205 ± 0.00757	5.71 ± 1.04
No echo	Paired	Reset	1.863 ± 0.031	0.85928 ± 0.00096	32.86 ± 2.15
No echo	Paired	Cumulative	1.770 ± 0.000	0.86486 ± 0.00000	100.00 ± 0.00
No echo	Paired	Window 8	1.770 ± 0.000	0.86486 ± 0.00000	100.00 ± 0.00
No echo	Randomized	Reset	1.947 ± 0.042	0.85773 ± 0.00129	50.15 ± 2.11
No echo	Randomized	Cumulative	1.770 ± 0.000	0.86349 ± 0.00031	59.33 ± 7.63
No echo	Randomized	Window 8	1.770 ± 0.000	0.86324 ± 0.00021	53.47 ± 5.24
No echo	Omit echo	Reset	1.861 ± 0.031	0.86206 ± 0.00096	98.34 ± 0.57
No echo	Omit echo	Cumulative	1.770 ± 0.000	0.86486 ± 0.00000	100.00 ± 0.00
No echo	Omit echo	Window 8	1.770 ± 0.000	0.86486 ± 0.00000	100.00 ± 0.00
No echo	Fixed exposed	Fixed	3.520 ± 0.000	0.87440 ± 0.00000	0.00 ± 0.00
No echo	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
High echo	Final only	Reset	10.631 ± 0.100	0.62109 ± 0.00317	0.00 ± 0.00
High echo	Final only	Cumulative	3.308 ± 0.294	0.82076 ± 0.00821	21.53 ± 6.14
High echo	Final only	Window 8	9.179 ± 0.191	0.65758 ± 0.00552	5.08 ± 1.09
High echo	Paired	Reset	1.895 ± 0.037	0.85708 ± 0.00111	97.61 ± 0.68
High echo	Paired	Cumulative	1.770 ± 0.000	0.86086 ± 0.00000	100.00 ± 0.00
High echo	Paired	Window 8	1.770 ± 0.000	0.86086 ± 0.00000	100.00 ± 0.00
High echo	Randomized	Reset	2.263 ± 0.071	0.84645 ± 0.00212	92.68 ± 1.01
High echo	Randomized	Cumulative	1.770 ± 0.000	0.86113 ± 0.00000	100.00 ± 0.00
High echo	Randomized	Window 8	1.770 ± 0.000	0.86113 ± 0.00000	100.00 ± 0.00
High echo	Omit echo	Reset	9.977 ± 0.049	0.61864 ± 0.00147	0.00 ± 0.00
High echo	Omit echo	Cumulative	9.821 ± 0.000	0.62334 ± 0.00000	0.00 ± 0.00
High echo	Omit echo	Window 8	9.821 ± 0.000	0.62334 ± 0.00000	0.00 ± 0.00
High echo	Fixed exposed	Fixed	9.637 ± 0.000	0.69089 ± 0.00000	0.00 ± 0.00
High echo	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
Both dependences	Final only	Reset	10.605 ± 0.090	0.62261 ± 0.00284	0.00 ± 0.00
Both dependences	Final only	Cumulative	9.145 ± 0.183	0.65054 ± 0.00502	8.01 ± 4.00
Both dependences	Final only	Window 8	10.305 ± 0.142	0.62560 ± 0.00439	0.98 ± 0.44
Both dependences	Paired	Reset	7.830 ± 0.030	0.68209 ± 0.00064	51.46 ± 2.24
Both dependences	Paired	Cumulative	7.297 ± 0.047	0.69368 ± 0.00102	92.19 ± 3.82
Both dependences	Paired	Window 8	7.580 ± 0.056	0.68748 ± 0.00119	69.04 ± 4.45
Both dependences	Randomized	Reset	8.269 ± 0.078	0.66957 ± 0.00222	47.17 ± 2.10
Both dependences	Randomized	Cumulative	7.372 ± 0.065	0.69228 ± 0.00141	85.94 ± 5.29
Both dependences	Randomized	Window 8	7.605 ± 0.056	0.68720 ± 0.00119	67.09 ± 4.45
Both dependences	Omit echo	Reset	13.735 ± 0.170	0.50836 ± 0.00526	0.00 ± 0.00
Both dependences	Omit echo	Cumulative	16.695 ± 0.321	0.41632 ± 0.00971	0.00 ± 0.00
Both dependences	Omit echo	Window 8	15.002 ± 0.339	0.46839 ± 0.01040	0.00 ± 0.00
Both dependences	Fixed exposed	Fixed	9.919 ± 0.000	0.68242 ± 0.00000	0.00 ± 0.00
Both dependences	Fixed blind	Fixed	8.560 ± 0.000	0.72020 ± 0.00000	0.00 ± 0.00
Echo shift	Final only	Reset	10.644 ± 0.094	0.62065 ± 0.00299	0.00 ± 0.00
Echo shift	Final only	Cumulative	4.285 ± 0.411	0.79352 ± 0.01167	24.41 ± 6.42
Echo shift	Final only	Window 8	9.043 ± 0.201	0.66161 ± 0.00589	6.40 ± 1.25
Echo shift	Paired	Reset	1.890 ± 0.034	0.85722 ± 0.00104	97.80 ± 0.63
Echo shift	Paired	Cumulative	1.770 ± 0.000	0.86086 ± 0.00000	100.00 ± 0.00
Echo shift	Paired	Window 8	1.770 ± 0.000	0.86086 ± 0.00000	100.00 ± 0.00
Echo shift	Randomized	Reset	2.321 ± 0.080	0.84473 ± 0.00238	91.89 ± 1.12
Echo shift	Randomized	Cumulative	1.837 ± 0.057	0.85915 ± 0.00169	99.17 ± 0.71
Echo shift	Randomized	Window 8	1.770 ± 0.000	0.86113 ± 0.00000	100.00 ± 0.00
Echo shift	Omit echo	Reset	9.981 ± 0.048	0.61851 ± 0.00144	0.00 ± 0.00
Echo shift	Omit echo	Cumulative	9.821 ± 0.000	0.62334 ± 0.00000	0.00 ± 0.00
Echo shift	Omit echo	Window 8	9.821 ± 0.000	0.62334 ± 0.00000	0.00 ± 0.00

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Environment	Policy	Memory	Late error (%)	Late net	Late hit (%)
Echo shift	Fixed exposed	Fixed	9.637 ± 0.000	0.69089 ± 0.00000	0.00 ± 0.00
Echo shift	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
Source asymmetry	Final only	Reset	10.490 ± 0.088	0.62523 ± 0.00279	0.00 ± 0.00
Source asymmetry	Final only	Cumulative	4.391 ± 0.239	0.79002 ± 0.00652	12.65 ± 4.99
Source asymmetry	Final only	Window 8	8.316 ± 0.244	0.68341 ± 0.00695	4.30 ± 1.01
Source asymmetry	Paired	Reset	2.964 ± 0.052	0.82508 ± 0.00158	90.92 ± 1.38
Source asymmetry	Paired	Cumulative	2.685 ± 0.000	0.83342 ± 0.00000	100.00 ± 0.00
Source asymmetry	Paired	Window 8	2.685 ± 0.000	0.83342 ± 0.00000	100.00 ± 0.00
Source asymmetry	Randomized	Reset	3.393 ± 0.054	0.81400 ± 0.00163	49.02 ± 2.20
Source asymmetry	Randomized	Cumulative	2.887 ± 0.055	0.82859 ± 0.00139	75.83 ± 6.61
Source asymmetry	Randomized	Window 8	3.004 ± 0.043	0.82562 ± 0.00107	61.77 ± 5.09
Source asymmetry	Omit echo	Reset	4.310 ± 0.173	0.78864 ± 0.00519	0.00 ± 0.00
Source asymmetry	Omit echo	Cumulative	3.521 ± 0.000	0.81234 ± 0.00000	0.00 ± 0.00
Source asymmetry	Omit echo	Window 8	3.521 ± 0.000	0.81234 ± 0.00000	0.00 ± 0.00
Source asymmetry	Fixed exposed	Fixed	5.101 ± 0.000	0.82697 ± 0.00000	0.00 ± 0.00
Source asymmetry	Fixed blind	Fixed	4.240 ± 0.000	0.84980 ± 0.00000	0.00 ± 0.00
Stronger humans	Final only	Reset	18.684 ± 0.136	0.37746 ± 0.00385	0.00 ± 0.00
Stronger humans	Final only	Cumulative	7.391 ± 0.370	0.69146 ± 0.01067	15.82 ± 5.67
Stronger humans	Final only	Window 8	12.709 ± 0.342	0.54463 ± 0.00987	3.61 ± 1.19
Stronger humans	Paired	Reset	5.481 ± 0.079	0.74670 ± 0.00243	75.88 ± 2.18
Stronger humans	Paired	Cumulative	4.600 ± 0.000	0.77377 ± 0.00000	100.00 ± 0.00
Stronger humans	Paired	Window 8	4.615 ± 0.017	0.77329 ± 0.00052	99.56 ± 0.45
Stronger humans	Randomized	Reset	5.591 ± 0.079	0.74361 ± 0.00240	73.10 ± 2.05
Stronger humans	Randomized	Cumulative	4.600 ± 0.000	0.77404 ± 0.00000	100.00 ± 0.00
Stronger humans	Randomized	Window 8	4.613 ± 0.013	0.77362 ± 0.00041	99.61 ± 0.38
Stronger humans	Omit echo	Reset	10.863 ± 0.191	0.59001 ± 0.00574	0.00 ± 0.00
Stronger humans	Omit echo	Cumulative	8.920 ± 0.000	0.64933 ± 0.00000	0.00 ± 0.00
Stronger humans	Omit echo	Window 8	8.952 ± 0.043	0.64831 ± 0.00129	0.00 ± 0.00
Stronger humans	Fixed exposed	Fixed	19.570 ± 0.000	0.39290 ± 0.00000	0.00 ± 0.00
Stronger humans	Fixed blind	Fixed	12.320 ± 0.000	0.60740 ± 0.00000	0.00 ± 0.00

D. Final-16-round costs and regret

Environment	Policy	Memory	Late acquisition	Late production	Late regret
No echo	Final only	Reset	215.187 ± 0.137	0.4351 ± 0.0094	0.24393 ± 0.00303
No echo	Final only	Cumulative	215.187 ± 0.137	1.1752 ± 0.0603	0.03929 ± 0.00639
No echo	Final only	Window 8	215.187 ± 0.137	0.8326 ± 0.0325	0.15724 ± 0.00757
No echo	Paired	Reset	233.600 ± 0.000	1.3864 ± 0.0044	0.00552 ± 0.00096
No echo	Paired	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.00000 ± 0.00000
No echo	Paired	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.00000 ± 0.00000
No echo	Randomized	Reset	232.500 ± 0.000	1.3511 ± 0.0048	0.00733 ± 0.00129
No echo	Randomized	Cumulative	232.500 ± 0.000	1.3313 ± 0.0153	0.00163 ± 0.00031
No echo	Randomized	Window 8	232.500 ± 0.000	1.3431 ± 0.0105	0.00186 ± 0.00021
No echo	Omit echo	Reset	233.600 ± 0.000	1.2543 ± 0.0015	0.00279 ± 0.00096
No echo	Omit echo	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.00000 ± 0.00000
No echo	Omit echo	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.00000 ± 0.00000
No echo	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.04749 ± 0.00000
No echo	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.05049 ± 0.00000
High echo	Final only	Reset	209.793 ± 0.110	0.4320 ± 0.0089	0.24546 ± 0.00317
High echo	Final only	Cumulative	209.793 ± 0.110	1.4389 ± 0.0728	0.04590 ± 0.00821
High echo	Final only	Window 8	209.793 ± 0.110	0.7887 ± 0.0335	0.20904 ± 0.00553
High echo	Paired	Reset	233.600 ± 0.000	1.4516 ± 0.0013	0.00377 ± 0.00111
High echo	Paired	Cumulative	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
High echo	Paired	Window 8	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
High echo	Randomized	Reset	232.500 ± 0.000	1.4446 ± 0.0025	0.01466 ± 0.00212
High echo	Randomized	Cumulative	232.500 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
High echo	Randomized	Window 8	232.500 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
High echo	Omit echo	Reset	233.600 ± 0.000	1.2506 ± 0.0008	0.24221 ± 0.00147
High echo	Omit echo	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.23752 ± 0.00000
High echo	Omit echo	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.23752 ± 0.00000

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Environment	Policy	Memory	Late acquisition	Late production	Late regret
High echo	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.22700 ± 0.00000
High echo	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.04649 ± 0.00000
Both dependences	Final only	Reset	206.910 ± 0.106	0.4305 ± 0.0082	0.07956 ± 0.00284
Both dependences	Final only	Cumulative	206.910 ± 0.106	1.2295 ± 0.0790	0.05173 ± 0.00502
Both dependences	Final only	Window 8	206.910 ± 0.106	0.7337 ± 0.0334	0.07663 ± 0.00439
Both dependences	Paired	Reset	233.600 ± 0.000	1.2952 ± 0.0152	0.01360 ± 0.00063
Both dependences	Paired	Cumulative	233.600 ± 0.000	1.5183 ± 0.0217	0.00209 ± 0.00102
Both dependences	Paired	Window 8	233.600 ± 0.000	1.4029 ± 0.0269	0.00827 ± 0.00119
Both dependences	Randomized	Reset	232.500 ± 0.000	1.2762 ± 0.0154	0.02638 ± 0.00222
Both dependences	Randomized	Cumulative	232.500 ± 0.000	1.4903 ± 0.0303	0.00376 ± 0.00141
Both dependences	Randomized	Window 8	232.500 ± 0.000	1.3928 ± 0.0275	0.00881 ± 0.00119
Both dependences	Omit echo	Reset	233.600 ± 0.000	1.1245 ± 0.0118	0.18733 ± 0.00527
Both dependences	Omit echo	Cumulative	233.600 ± 0.000	1.2902 ± 0.0084	0.27945 ± 0.00972
Both dependences	Omit echo	Window 8	233.600 ± 0.000	1.2252 ± 0.0164	0.22736 ± 0.01040
Both dependences	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.07038 ± 0.00000
Both dependences	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.03260 ± 0.00000
Echo shift	Final only	Reset	209.670 ± 0.124	0.4359 ± 0.0092	0.24594 ± 0.00299
Echo shift	Final only	Cumulative	209.670 ± 0.124	1.3365 ± 0.0699	0.07317 ± 0.01168
Echo shift	Final only	Window 8	209.670 ± 0.124	0.7926 ± 0.0303	0.20503 ± 0.00589
Echo shift	Paired	Reset	233.600 ± 0.000	1.4524 ± 0.0007	0.00363 ± 0.00104
Echo shift	Paired	Cumulative	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
Echo shift	Paired	Window 8	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
Echo shift	Randomized	Reset	232.500 ± 0.000	1.4427 ± 0.0025	0.01638 ± 0.00238
Echo shift	Randomized	Cumulative	232.500 ± 0.000	1.4483 ± 0.0014	0.00197 ± 0.00169
Echo shift	Randomized	Window 8	232.500 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
Echo shift	Omit echo	Reset	233.600 ± 0.000	1.2511 ± 0.0003	0.24234 ± 0.00144
Echo shift	Omit echo	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.23752 ± 0.00000
Echo shift	Omit echo	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.23752 ± 0.00000
Echo shift	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.22700 ± 0.00000
Echo shift	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.04649 ± 0.00000
Source asymmetry	Final only	Reset	211.038 ± 0.117	0.4223 ± 0.0082	0.21360 ± 0.00278
Source asymmetry	Final only	Cumulative	211.038 ± 0.117	1.3368 ± 0.0714	0.04891 ± 0.00653
Source asymmetry	Final only	Window 8	211.038 ± 0.117	0.7773 ± 0.0383	0.15547 ± 0.00696
Source asymmetry	Paired	Reset	233.600 ± 0.000	1.4478 ± 0.0025	0.00832 ± 0.00158
Source asymmetry	Paired	Cumulative	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
Source asymmetry	Paired	Window 8	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
Source asymmetry	Randomized	Reset	232.500 ± 0.000	1.3683 ± 0.0051	0.01962 ± 0.00163
Source asymmetry	Randomized	Cumulative	232.500 ± 0.000	1.4017 ± 0.0132	0.00510 ± 0.00139
Source asymmetry	Randomized	Window 8	232.500 ± 0.000	1.3735 ± 0.0102	0.00806 ± 0.00107
Source asymmetry	Omit echo	Reset	233.600 ± 0.000	1.2506 ± 0.0009	0.04477 ± 0.00519
Source asymmetry	Omit echo	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.02108 ± 0.00000
Source asymmetry	Omit echo	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.02108 ± 0.00000
Source asymmetry	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.06349 ± 0.00000
Source asymmetry	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.04066 ± 0.00000
Stronger humans	Final only	Reset	211.641 ± 0.115	0.5113 ± 0.0122	0.40155 ± 0.00386
Stronger humans	Final only	Cumulative	211.641 ± 0.115	1.7568 ± 0.0648	0.08766 ± 0.01067
Stronger humans	Final only	Window 8	211.641 ± 0.115	1.1179 ± 0.0290	0.23444 ± 0.00987
Stronger humans	Paired	Reset	233.600 ± 0.000	1.5895 ± 0.0085	0.02702 ± 0.00243
Stronger humans	Paired	Cumulative	233.600 ± 0.000	1.5600 ± 0.0000	0.00000 ± 0.00000
Stronger humans	Paired	Window 8	233.600 ± 0.000	1.5610 ± 0.0016	0.00048 ± 0.00052
Stronger humans	Randomized	Reset	232.500 ± 0.000	1.5928 ± 0.0076	0.03037 ± 0.00239
Stronger humans	Randomized	Cumulative	232.500 ± 0.000	1.5600 ± 0.0000	0.00000 ± 0.00000
Stronger humans	Randomized	Window 8	232.500 ± 0.000	1.5610 ± 0.0014	0.00042 ± 0.00041
Stronger humans	Omit echo	Reset	233.600 ± 0.000	1.3511 ± 0.0098	0.18371 ± 0.00574
Stronger humans	Omit echo	Cumulative	233.600 ± 0.000	1.3020 ± 0.0000	0.12444 ± 0.00000
Stronger humans	Omit echo	Window 8	233.600 ± 0.000	1.3053 ± 0.0030	0.12545 ± 0.00129
Stronger humans	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.43790 ± 0.00000
Stronger humans	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.22340 ± 0.00000

E. Paired policy and memory contrasts

Environment	Policy A / memory	Policy B / memory	Error diff. (pp)	Net difference	Regret diff.
No echo	Paired / Cumulative	Omit echo / Cumulative	0.000 ± 0.000	-0.00005 ± 0.00001	0.00005 ± 0.00001
No echo	Paired / Cumulative	Randomized / Cumulative	0.000 ± 0.004	0.00142 ± 0.00027	-0.00168 ± 0.00027
No echo	Paired / Window 8	Paired / Cumulative	0.000 ± 0.000	0.00000 ± 0.00000	0.00000 ± 0.00000
No echo	Randomized / Window 8	Randomized / Cumulative	0.000 ± 0.000	-0.00012 ± 0.00017	0.00011 ± 0.00017
No echo	Paired / Cumulative	Fixed blind / Fixed	-1.748 ± 0.002	-0.00665 ± 0.00008	-0.05038 ± 0.00008
High echo	Paired / Cumulative	Omit echo / Cumulative	-8.052 ± 0.001	0.23755 ± 0.00004	-0.23755 ± 0.00004
High echo	Paired / Cumulative	Randomized / Cumulative	-0.009 ± 0.008	-0.00000 ± 0.00025	-0.00027 ± 0.00025
High echo	Paired / Window 8	Paired / Cumulative	0.000 ± 0.000	0.00000 ± 0.00000	0.00000 ± 0.00000
High echo	Randomized / Window 8	Randomized / Cumulative	0.000 ± 0.000	0.00000 ± 0.00000	0.00000 ± 0.00000
High echo	Paired / Cumulative	Fixed blind / Fixed	-1.747 ± 0.003	-0.01062 ± 0.00009	-0.04641 ± 0.00009
Both dependences	Paired / Cumulative	Omit echo / Cumulative	-8.516 ± 0.365	0.25134 ± 0.01073	-0.25134 ± 0.01073
Both dependences	Paired / Cumulative	Randomized / Cumulative	-0.075 ± 0.061	0.00154 ± 0.00131	-0.00181 ± 0.00131
Both dependences	Paired / Window 8	Paired / Cumulative	0.140 ± 0.034	-0.00310 ± 0.00073	0.00309 ± 0.00073
Both dependences	Randomized / Window 8	Randomized / Cumulative	0.132 ± 0.040	-0.00295 ± 0.00084	0.00293 ± 0.00084
Both dependences	Paired / Cumulative	Fixed blind / Fixed	-1.137 ± 0.049	-0.02921 ± 0.00105	-0.02783 ± 0.00105
Echo shift	Paired / Cumulative	Omit echo / Cumulative	-4.019 ± 0.006	0.11851 ± 0.00017	-0.11851 ± 0.00017
Echo shift	Paired / Cumulative	Randomized / Cumulative	-0.334 ± 0.084	0.01049 ± 0.00239	-0.01075 ± 0.00239
Echo shift	Paired / Window 8	Paired / Cumulative	-0.007 ± 0.006	0.00019 ± 0.00017	-0.00019 ± 0.00017
Echo shift	Randomized / Window 8	Randomized / Cumulative	-0.153 ± 0.076	0.00453 ± 0.00224	-0.00453 ± 0.00224
Echo shift	Paired / Cumulative	Fixed blind / Fixed	-1.741 ± 0.006	-0.00885 ± 0.00018	-0.04819 ± 0.00018
Source asymmetry	Paired / Cumulative	Omit echo / Cumulative	-0.853 ± 0.010	0.02158 ± 0.00031	-0.02158 ± 0.00031
Source asymmetry	Paired / Cumulative	Randomized / Cumulative	-0.244 ± 0.046	0.00590 ± 0.00116	-0.00616 ± 0.00116
Source asymmetry	Paired / Window 8	Paired / Cumulative	0.000 ± 0.000	0.00000 ± 0.00000	0.00000 ± 0.00000
Source asymmetry	Randomized / Window 8	Randomized / Cumulative	0.072 ± 0.030	-0.00183 ± 0.00075	0.00182 ± 0.00074
Source asymmetry	Paired / Cumulative	Fixed blind / Fixed	-1.547 ± 0.004	-0.01662 ± 0.00013	-0.04041 ± 0.00013
Stronger humans	Paired / Cumulative	Omit echo / Cumulative	-4.347 ± 0.016	0.12525 ± 0.00048	-0.12525 ± 0.00048
Stronger humans	Paired / Cumulative	Randomized / Cumulative	-0.016 ± 0.013	0.00021 ± 0.00039	-0.00048 ± 0.00039
Stronger humans	Paired / Window 8	Paired / Cumulative	0.006 ± 0.006	-0.00018 ± 0.00018	0.00018 ± 0.00018
Stronger humans	Randomized / Window 8	Randomized / Cumulative	0.007 ± 0.006	-0.00024 ± 0.00019	0.00024 ± 0.00019
Stronger humans	Paired / Cumulative	Fixed blind / Fixed	-7.697 ± 0.009	0.16567 ± 0.00028	-0.22270 ± 0.00028